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Children's comprehension and repair of garden-path *wh*-questions

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ABSTRACT

Children's sentence interpretations often lack flexibility. For example, when French-speaking adults and children hear ambiguous *wh*-questions like *Where did Annie explain that she rode her horse?*, they preferentially associate the *wh*-phrase with the first verb and adopt the main clause interpretation (e.g., *She explained at the campsite*). This preferred association results in shorter syntactic dependencies compared to second verb associations (i.e., the embedded clause interpretation: *She rode in the forest*). Moreover, this bias toward shorter dependencies persists in "filled-gap" *wh*-questions, where the preferred interpretation is blocked by a prepositional phrase (e.g., *Where did Annie explain **at the campsite** that she rode her horse?*). Here, adults preferentially suppress their main clause bias and adopt embedded clause interpretations, whereas children still prefer their initial interpretations. The present study investigates how English-speaking children interpret ambiguous and filled-gap *wh*-questions. Cross-linguistic evidence suggests that children struggle to adopt second verb associations when first verb interpretations are unavailable. In five story-based experiments, we show that English-speaking children prefer main clause (first verb) interpretations in both *wh*-question conditions—although they seem to adopt more second verb associations than has been previously reported for children in other languages. We also document a novel repair strategy used by adults and children in which they describe *who* was at the main clause event rather than *where* it occurred. Taken together, our findings highlight the cross-linguistic stability of children's shorter dependency biases and suggest that children's abilities to inhibit preferred interpretations may be shaped by the development of language-specific syntactic knowledge.

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1. Introduction

Much language development research has focused on the acquisition of linguistic structures to understand how learners discover the abstract structural representations through input signals in the environment. However, another learning challenge that has received relatively less attention in the literature is how children learn to parse those structures in an efficient, adult-like fashion. Just like linguists have uncovered grammatical variations across languages, adult sentence processing research has demonstrated that parsing biases can vary across languages too—potentially as a result of adaptation to language-specific properties and distributions of syntactic structures (e.g., Cuetos, Mitchell & Martin 1996). Developmental psycholinguistics research has also revealed that children's language processing behaviors often diverge from those of adults despite their sophisticated linguistic knowledge (e.g., Choi & Trueswell 2010; Lidz, Gleitman & Gleitman 2003; Trueswell et al. 1999). However, it is still largely unknown how or when children acquire adult-like parsing strategies, as children cannot directly inspect how adults parse their linguistic input. All of these observations indicate that parser development poses a very similar learning problem to that of the acquisition of

linguistic structures: How do children learn the target linguistic system in the absence of direct access to adults' mental operations? As such, understanding the process of parser development could serve as a potential source of insight on the learning mechanism or innate predisposition that enables the feat of language development more generally.

The present article sheds light on this question by investigating English-speaking 5-year-old children's ability to interpret garden-path *wh*-questions like *Where did Annie explain at the campsite that she rode her horse?* In these temporarily ambiguous questions, the early main clause portion of the sentence could be initially construed as a question about the location of the *explaining* event. However, information immediately after the verb (e.g., *at the campsite*) conflicts with this analysis and instead signals that the question is about the location of Annie's *horse riding*. This sentence demonstrates the typical characteristics of syntactic environments in which children incrementally assign an interpretation to the partial input and then fail to revise their initial interpretive commitments, such that they end up with non-adult-like interpretations (e.g., Trueswell et al. 1999).

The first part of this study builds on previous work in Japanese and French, which shows that children have a general bias toward shorter dependencies called *active gap-filling* (Stowe 1986). In five story-based experiments, we show that English-speaking children have a similar bias to associate the *wh*-phrase with the main clause (first verb) even if that interpretation is syntactically unavailable (i.e., in garden-path *wh*-questions). However, the current work goes beyond the previous studies by showing that English-speaking children often suppress this bias and adopt the embedded clause (second verb) interpretation at a higher rate than has been previously reported for children in the cross-linguistic literature.

The second part of this study investigates a unique strategy adopted by English speakers to repair garden-path *wh*-questions. More specifically, we observed a novel strategy in which adults and children described *who* was at the main clause event (e.g., *her friend*) rather than *where* the event occurred. We argue that this "*who* substitution" strategy arises from pragmatic and discourse factors in our design but nonetheless reflects the strength of participants' active gap-filling bias. Taken together, the present study illustrates the cross-linguistic stability of listeners' biases toward shorter dependencies and suggests that the strength of their biases may be shaped by language-specific properties surrounding the formation of *wh*-questions.

2. Background on children's interpretation of garden-path *wh*-questions

One robust finding in child sentence processing research is that children's sentence interpretations often lack flexibility. One of the strongest pieces of evidence for this generalization comes from 5-year-old children's comprehension of act-out instructions involving temporary Prepositional Phrase (PP) attachment ambiguities, such as *Put the frog on the napkin in the box*. When the test scene contained an empty napkin, an empty box, and a frog that was sitting on a napkin, eye movement data indicated that both adults and children incrementally interpreted the PP (*on the napkin*) as the destination for moving the frog. However, subsequent actions revealed a contrast between adults' and children's ultimate interpretations of the sentence. While adults revised the interpretation of *on the napkin* as a modifier of *the frog* and analyzed *in the box* as the destination, 5-year-old children often interpreted the napkin as the destination (Hurewitz et al. 2000; Trueswell et al. 1999; Weighall 2008; for related findings in Korean, see Choi & Trueswell 2010). These findings suggest that the errors in children's final interpretations may stem from difficulties with revising their initial analysis of the PP attachment ambiguity.

Subsequent work has found converging evidence for sentence revision difficulties in other (temporarily) ambiguous sentence environments, such as passives (Huang, Leech & Rowe 2017; Huang et al. 2013), quantifier scope ambiguities (Musolino & Lidz 2006; Viau, Lidz & Musolino 2010; for a review, see Lidz 2018), and sentences with ambiguous words like polysemes and homophones (Rabagliati, Pylkkänen & Marcus 2013; for reviews on parser development, see Omaki & Lidz 2015; Snedeker & Huang 2015). The present article focuses on children's incremental processing of filler-gap dependencies in *wh*-questions (e.g., *Where did Annie explain that she rode her horse?*), which is one of the few structures that has been investigated across languages with different grammatical properties.

Filler-gap dependencies create a temporary structural ambiguity, as the left-dislocated constituent in a noncanonical structural position (the *filler*) needs to be associated with a late-arriving structural position where that constituent would typically appear (the *gap*). Adult psycholinguistics research has found a wide range of evidence for an active gap-filling bias, where adults incrementally assign the filler to the earliest, grammatically permissible gap position to shorten the dependency (e.g., Aoshima, Phillips & Weinberg 2004; Frazier 1987; Garnsey, Tanenhaus & Chapman 1989; Omaki et al. 2015; Omaki & Schulz 2011; Pickering & Traxler 2003; Staub 2007; Sussman & Sedivy 2003; Traxler & Pickering 1996). For example, the seminal study by Stowe (1986) used a so called *filled-gap* design (1) and found that the reading time for the direct object *us* was significantly higher in (1a) with a filler-gap dependency than in (1b) with an embedded polar question (i.e., no filler-gap dependency). This reading disruption in (1a) or the *filled-gap effect* suggests that readers incrementally analyzed the fronted *wh*-phrase *who* as the direct object of *bring*, but they had to revise this initial analysis when the actual direct object was presented.

- (1) a. My brother wanted to know **who** Ruth will bring **us** home to __ at Christmas.
b. My brother wanted to know **if** Ruth will bring **us** home to Mom at Christmas.

Child sentence processing research has found evidence of similar biases in 5- to 6-year-old children by studying their comprehension of *wh*-questions. For example, Omaki et al. (2014) presented adults and 5-year-old children with globally ambiguous *wh*-questions like (2) in which the answer could be about the *telling* location or the *butterfly-catching* location.

- (2) a. Where did Lizzie tell someone that she was gonna catch butterflies?
b. *Doko-de Yukiko-chan-wa choucho-o tsukamaeru-to itteta-no?*
where-at Yukiko-Dim-Top pro butterfly-Acc catch- Comp was telling-Q
'Where was Yukiko telling someone that she would catch butterflies?'

These questions were presented after short stories in which the prominence of main clause and embedded clause events was balanced. Both English-speaking adults and 5-year-old children showed a strong preference for the main clause (*telling* location) interpretation of (2a). Moreover, when Japanese-speaking adults and children were given the Japanese translations of the stories and ambiguous *wh*-questions like (2b), they showed an opposite bias for the embedded clause interpretation. While these findings may appear to be at odds with each other, together they constitute strong evidence for a bias toward shorter dependencies: Both English and Japanese speakers show the same preference to associate the *wh*-phrase with the first verb in the sentence, which turns out to be the main clause verb in English and the embedded clause verb in Japanese. Additional evidence for children's shorter dependency bias comes from French. For example, Lassotta, Omaki & Franck (2016) presented French-speaking adults and 5- to 8-year-old children with the French translations of (2a) and found a similar preference for main clause (first verb) interpretations of globally ambiguous *wh*-questions.

These French and Japanese studies also looked at *wh*-questions with a filled-gap design to gain further insight into children's parsing biases. Omaki and his colleagues used sentences like (3) to investigate whether children's previously observed biases persist in "filled-gap" garden-path *wh*-questions (for Japanese, see Omaki et al. 2014; for French, see Lassotta, Omaki & Franck 2016).

- (3) a. *Doko-de Yukiko-chan-wa kouen-de choucho-o tsukamaeru-to itteta-no?*
where-at Yukiko-Dim-Top pro park-at butterfly-Acc catch-Comp was telling-Q
'Where was Yukiko telling someone that she will catch butterflies at the park?'
b. *Où est-ce qu'Aline a expliqué dans le salon qu'elle allait attraper des papillons?*
where Q Aline has explained in the room that she was gonna catch some butterflies
'Where did Aline explain in the living room that she was going to catch butterflies?'

Here, the target sentences contained a PP that specified the location of the embedded clause event in Japanese (3a) or the main clause event in French (3b) such that the preferred answers in the ambiguous versions were now blocked. Omaki et al. (2014) found that Japanese children provided the (incorrect) embedded clause answer *kouen* ‘park’ to (3a) despite the fact that it is already mentioned in the question. Japanese-speaking children provided these (incorrect) embedded clause responses at a rate that was not reliably different from the rate of (acceptable) embedded clause responses in the ambiguous condition (embedded clause interpretation rate: 94% for ambiguous conditions, 83% for filled-gap conditions). Japanese-speaking adults performed as expected, providing more embedded clause responses in the ambiguous condition (92%) than in the filled-gap condition (5%). The results from Omaki et al. (2014) provide clear evidence that Japanese-speaking children prefer their initial interpretations (even in the face of contradictory information), whereas adults suppress their initial commitments and adopt the second verb interpretation of the filled-gap *wh*-question.

In French, however, the results are not as clear cut. Lassotta, Omaki & Franck (2016) found that French-speaking children behaved similarly to Japanese-speaking children, as they preferred main clause (first verb) interpretations for both ambiguous (85%) and filled-gap *wh*-questions (88%). Interestingly, French-speaking adults also preferred main clause interpretations in both *wh*-question conditions (ambiguous: 88%; filled-gap: 67%). Given this unexpected finding, Lassotta, Omaki & Franck (2016) conducted a follow-up acceptability judgment experiment with *wh*-questions like (3b). They found that French-speaking adults strongly disprefer the long-distance *wh*-questions with cross-clausal filler-gap dependencies and instead preferred the option of having an in-situ *wh*-phrase in the embedded clause (i.e., *Où* ‘where’ in (3b) would appear in sentence final position).¹ From this data pattern, Lassotta, Omaki & Franck (2016) argued that French-speaking adults have a general grammatical bias (on top of their active gap-filling bias) to associate the fronted *Où* ‘where’ in (3b) with the main clause—and that this additional bias may have inflated adults’ (incorrect) main clause preferences for garden-path *wh*-questions.

Lassotta, Omaki & Franck (2016) discuss this general grammatical bias as only affecting adults’ performance; however, it is entirely possible that the same bias inflated children’s main clause preferences too. We have suggested that language-specific properties may shape how listeners parse sentences in their language. If French speakers prefer to use in-situ *wh*-constructions when the embedded clause reading is intended, then the majority of linguistic input with fronted *wh*-phrases will be intended to have main clause interpretations—even though fronted *wh*-phrases with cross-clausal dependencies are permissible in French. Thus, there is a gap in the literature on children’s comprehension of garden-path *wh*-questions in a head-initial language that *clearly* allows for constructions like (3b).

3. The present study

It remains unclear the degree to which active gap-filling biases influence children’s comprehension of filled-gap *wh*-questions in a head-initial language like English, which clearly allows for cross-clausal dependencies with fronted *wh*-phrases (3b). The main goal of this study was to fill this gap in the literature and assess how English-speaking children interpreted garden-path *wh*-questions. A secondary goal was to see if children’s knowledge of language-specific properties regarding question formation (i.e., the optionality of *wh*-fronting in French and the obligatory *wh*-fronting in English) impacted how strongly speakers preferred first verb interpretations. For example, direct questions in English obligatorily require the *wh*-phrase to be fronted to the sentence-initial position. Without French-like optionality in question formation, English speakers may be better able to assign the correct, embedded clause interpretation to English garden-path *wh*-questions. Thus, we ask if English-

¹Interestingly, this finding is contrary to some claims in the syntax literature (e.g., Cheng & Rooryck 2000; for further experimental evidence for embedded *wh*-in-situ in French, see Oiry 2011).

speaking children behave any differently than French-speaking children with respect to their preferred interpretations of these garden-path *wh*-questions.

On the one hand, given the general sentence revision difficulties that are widely observed in children at this age, English-speaking 5-year-olds may struggle to assign the embedded clause interpretation to a similar degree as the French-speaking children. On the other hand, it is possible that French-speaking children's rate of main clause responses was inflated because of the same grammatical preference found in French-speaking adults; by extension, if 5-year-old English-speaking children already have the grammatical knowledge of obligatory *wh*-fronting in English, then English-speaking children may demonstrate a higher rate of embedded clause interpretations than French-speaking children. These questions, as well as questions regarding English speakers' strategies for interpreting garden-path *wh*-questions, will be investigated in five story-based comprehension experiments.

3.1. Experiment 1: Comprehension of ambiguous and filled-gap *wh*-questions in English

Experiment 1 investigates how English-speaking adults and children interpret globally ambiguous (4a) and filled-gap, bi-clausal *wh*-questions (4b) using a Question-after-Story paradigm (de Villiers, Roeper & Vainikka 1990).

- (4) a. Where did Annie *explain* that she rode her horse?
- b. Where did Annie *explain at the campsite* that she rode her horse?

In Experiment 1, both *wh*-questions are directly modeled after those used in the French study by Lassotta, Omaki & Franck (2016). Using *explain* as the main clause verb has two advantages: First, it was used in the original French study, which keeps the stimuli maximally similar and allows for stronger comparisons across the two languages.² Second, *explain* does not require an overt indirect object (i.e., the recipient of the *explaining* does not need to be mentioned). Without an additional noun phrase (NP) after the verb, we can ensure that the PP (*at the campsite*) attaches to the main clause verb in filled-gap conditions (*explaining at the campsite*). If there was an overt indirect object, the PP could be interpreted as attaching low and modifying the NP (e.g., *explain to* [someone at the campsite]_{NP}). If the PP has low attachment, it cannot properly block the main clause interpretation, leading to a grammatically permissible main clause interpretation of the garden-path *wh*-question.

3.1.1. Participants

We recruited 28 adult native English speakers from the community at Harvard University. Twenty-six children were also recruited for participation from the Greater Boston area. Of those participants, we excluded four adults and two children from further analyses for providing more than two incorrect or irrelevant answers out of the eight total filler and test questions. Adult participants ($N = 24$) ranged in age from 16–57 (mean age = 19.6) and were compensated at a 10 USD/hour rate or with course credit. The children in the ambiguous ($N = 12$) and filled-gap ($N = 12$) conditions ranged from 5;01 to 6;00 (mean age = 5;06) and 5;00 to 6;01 (mean age = 5;06) respectively.

3.1.2. Materials

3.1.2.1. Story design. Our task used the same eight stories as Omaki et al. (2014). These eight stories are kept the same throughout all five experiments reported in this article. Each story was created from sequences of clipart animations. In each story display, there were four images of locations that are familiar to children (e.g., *the school*, *the playground*), one in each quadrant (see Figure 1). A full example story with its target question(s) is listed in (5).

²Lassotta, Omaki & Franck (2016) used both *expliquer* 'explain' and *raconter* 'tell' as main clause verbs. Both verbs in French can occur without an indirect object. In English, however, only *explain* can occur without an indirect object.

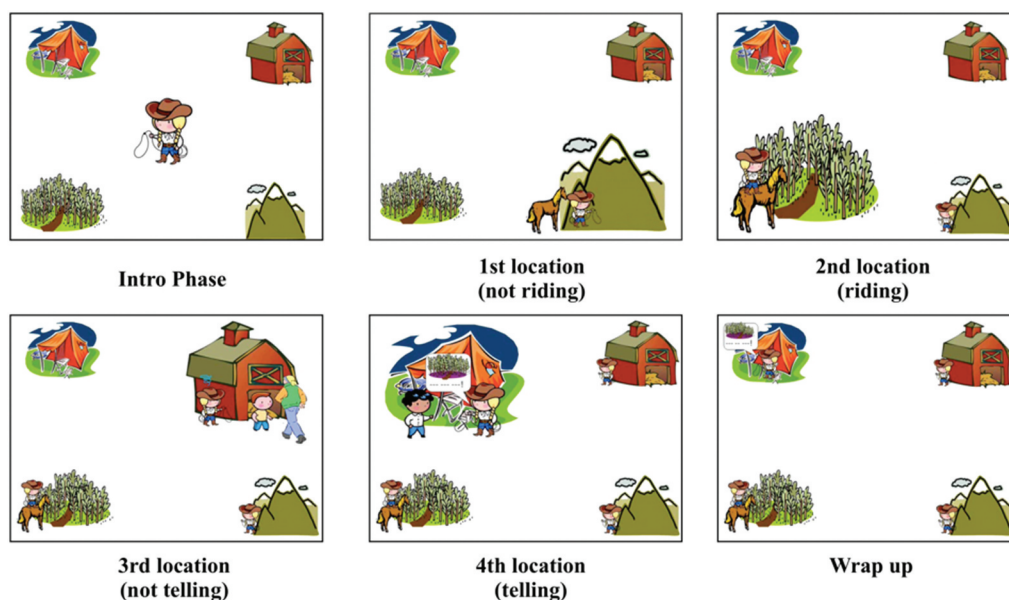


Figure 1. A sample sequence from the scenes used to accompany the stories in Experiments 1–5.

(5) *Full example of a story script for a given trial:*

[1st location] Annie was taking horseback riding lessons. Today she got to go on a long ride all by herself. First she tried to ride up a mountain, but the mountain trail was too steep for the horse so she couldn't ride the horse there at all.

[2nd location] Instead, Annie and her horse went riding in the forest. It was so much fun to ride the horse in the forest—along the way, Annie and her horse even got to jump over a log that was blocking the path! Annie was really proud of herself for riding all alone in the forest, so she wanted to tell her riding teacher about her great ride.

[3rd location] But when Annie arrived at the stable, her teacher was busy teaching another student, so she couldn't talk to her.

[4th location] That night, Annie and her friend went camping at a campsite. Annie's friend was taking horseback riding lessons too, so he asked Annie about her ride. Annie said to him, "I rode a horse in the forest today, and we jumped over a big log!" Annie's friend was jealous that she's already allowed to go riding by herself.

[Question] Where did Annie explain (at the campsite) that she rode her horse?

In each story, the main character visits four distinct locations. Each location was magnified when the character arrived in the story to highlight the relevant location and make it clearly visible to participants. Each story trial had six critical phases: (i) *the introductory phase*, where the main character began centered in the display and participants learned more about the theme of the story; (ii–v) *the visiting phases*, where the main character visited each location and either failed or succeeded in doing an activity relevant to the story; and (vi) *the final wrap-up phase*, where the character completed the intended activity and the location shrank to its original size. After this phase, the target question was presented auditorily by the experimenter. All study materials can be found on OSF (osf.io/qc84n/).

At each location, the character attempted a particular task relevant to the story's theme. In each story, there were two locations related to the main clause event and two related to the embedded clause event. At the first location, the particular event would almost occur but eventually fail; only at the second location would the character actually succeed. In the previous example, Annie goes to

the first location (*the mountain*) to participate in the embedded clause event (*horse riding*) but fails. She then goes to the second location (*the forest*) and succeeds at riding her horse. Similarly, for the main clause event (*explaining*), she goes to the first location (*the stable*) to talk about her ride. Here, she almost succeeds in explaining the ride to her instructor, but she ultimately fails. Only at the second location (*the campsite*) was she able to successfully explain to her friend about her ride. The motivation for using four locations was to make the *where* target questions felicitous and to make each location-event pairing more interesting and memorable for children. This four-location design is identical to Omaki et al. (2014), but it differs slightly from Lassotta, Omaki & Franck (2016), which only used three locations: one for the main clause event, one for the embedded clause, and a third distractor scene. We discuss these design differences and how they could potentially influence participants' response strategies in the Discussion.

Finally, half of the stories first presented locations relevant to the embedded clause event, while the other half first presented those relevant to the main clause event. This design controlled for potential recency biases in answers, i.e., participants providing answers consistent with the last event presented in the story. Regardless of event order, the first sequence of events always motivated the subsequent set of events. We also left visual traces to help children remember what happened in each location. For example, at successful locations, a visual trace of that event would remain on the display (e.g., Annie on her horse or Annie talking with a speech bubble), but at failed locations only the character image would remain. Each quadrant of the visual display depicted the relevant events randomly across stories to control for quadrant biases and predictions of event type based on image position.

3.1.2.2. Question design. All eight stories were paired with a comprehension *wh*-question. Four of the stories served as targets with bi-clausal *where* questions like (4), while the remaining four stories served as fillers with *why* questions to check the participants' attentiveness. None of the *why* questions directly probed location information nor referenced the successful events that happened at their respective locations. Most questions targeted the reason why some activity failed (e.g., *Why couldn't John talk to his mom?*). The *why* questions were often shorter and syntactically less complex than the *where* questions; however, we believe the *why* questions were slightly more challenging than the *where* questions because participants needed to recall more story details (without helpful visual cues) to provide a proper explanation (e.g., Q: *Why did Hannah leave the cave right away?* A: *Because the cave was too dark, and she got scared*; see de Villiers et al. 2008 for further discussion). Across all experiments, children's accuracy on these *why* filler questions ranged from 91.7% to 100%, and adults' accuracy ranged from 89.6% to 100%, suggesting a high level of engagement and story comprehension.

The target questions consisted of two between-participant conditions: the globally ambiguous (4a) and the filled-gap versions (4b). As previously mentioned, these questions were directly modeled after Lassotta, Omaki & Franck (2016). All target questions used *explain* as the main clause verb without an indirect object. This allowed us to avoid issues with interpreting the attachment of the PP in the filled-gap conditions (4b). Regardless of condition, all target questions contained the overt complementizer *that*, as pilot work by Omaki et al. (2014) suggested that the overt complementizer allows main clause and embedded clause interpretations to be equally accessible to adult English speakers. The embedded clause verb phrase (VP) region varied depending on the content of the character's speech: when a character talked about his or her plan to do something, the embedded clause used *he/she was gonna VERB*. However, when a character talked about a past event, this region used a simple past tense form of the verb. The eight stories were distributed across two lists to control for any order effects, trial saliency, or item-based interpretation biases. A list of story scripts and questions used in all five experiments can be found in the Appendix.

3.1.3. Procedure

Both adults and children received the same instructions from the experimenter as to minimize the development of different task strategies across populations. Participants were told that they were going to play a game with a puppet in which they would watch some short videos and answer questions

about what they saw. All participants saw two short practice trials before beginning the testing phase. The experimenter ensured that they understood the task and gave opportunities to ask any unanswered questions.

After the practice phase, the eight stories were presented in either List 1 (forward) or List 2 (reverse) order. The cartoon stories were presented using Microsoft PowerPoint and a set of USB speakers attached to a MacBook Pro laptop. The experimenter paused after each story and asked the target or filler question using a puppet. The experimenter was trained to produce the questions with neutral prosody that would not bias the listeners to one interpretation over another. For example, the prosodic prominence on the main clause VP region and embedded clause VP region was controlled to be comparable, as an imbalance in prosodic prominence could potentially affect the *wh*-phrase interpretation. The experimenter also made an effort to use comparable speech rates for the question presentation across trials. In Experiment 1, the experimenter presented ambiguous target questions with mean durations of 5.91 ($SE = 0.39$) and 5.53 ($SE = 0.27$) seconds for adults and children respectively. In the filled-gap condition, the experimenter had mean target durations of 7.46 ($SE = 0.26$) and 7.32 ($SE = 0.25$) seconds for adults and children respectively. Note that the durations of the filled-gap condition are greater than the ambiguous condition due to the additional PP in the question (e.g., *at the campsite*). All testing sessions were recorded on a digital voice recorder for subsequent data coding purposes. Each experiment took approximately 15 minutes, and all participants were able to finish the entire task.

3.1.4. Data coding

Participants' responses were transcribed and classified into five categories: (i) *main clause locations*; (ii) *sublocations*, in which participants provided a particular spot within the larger main clause location, e.g., *next to the tent* (at the larger campsite). These sublocations were not originally mentioned in the story and were entirely self-generated by participants. Sublocations are possible for both main clause and embedded clause locations, e.g., *on the path* (in the forest where Annie rode her horse); however, participants only provided main clause sublocations; (iii) *who substitution* responses, in which participants provided the recipient of the *explaining/telling* event (e.g., *her friend*) as if the *wh*-phrase was *who* rather than *where*. Interestingly, both adults and children provided this novel response type; (iv) *embedded clause locations*; and (v) *other*, including failure to provide any answer or one that did not correspond to the relevant main clause nor embedded clause events.

- (i) *main clause locations*;
- (ii) *sublocations*, in which participants provided a particular spot within the larger main clause location, e.g., *next to the tent* (at the larger campsite). These sublocations were not originally mentioned in the story and were entirely self-generated by participants. Sublocations are

Table 1. Response types for where did Annie explain (at the campsite) that she rode her horse?

Response type	Example	Description
1. main clause location	<i>at the campsite</i>	The location of the telling event
2. sublocation	<i>by the tent</i>	A subpart of the larger, main clause event location (e.g., the tent was inside the campsite) ^a
3. <i>who</i> substitution	<i>(to/with) her friend^b</i>	The recipient of the telling event
4. embedded clause location	<i>in the forest</i>	The location of the horse-riding event
5. other	<i>because it was fun</i>	Failure to respond or an irrelevant response

^aSublocation responses are possible for embedded clause events, e.g., **Q:** *Where did Annie explain that she rode her horse?* **A:** *On the path (in the forest)*. However, no participant provided sublocations for the embedded clause event.

^bParticipants varied in providing a preposition with the indirect object of the telling event. Instances including a preposition seem to strongly indicate a location, e.g., *with her friends is where she did the explaining*.

- possible for both main clause and embedded clause locations, e.g., *on the path* (in the forest where Annie rode her horse); however, participants only provided main clause sublocations;
- (iii) *who substitution* responses, in which participants provided the recipient of the *explaining/telling* event (e.g., *her friend*) as if the *wh*-phrase was *who* rather than *where*. Interestingly, both adults and children provided this novel response type;
 - (iv) *embedded clause locations*;
 - (v) *other*, including failure to provide any answer or one that did not correspond to the relevant main clause nor embedded clause events.

This data coding was used in all five experiments. Examples of these response types are illustrated in Table 1.

Three of these possible response types constitute a main clause interpretation of the filler-gap dependency: *main clause locations*, *sublocations*, and *who substitutions*. We argue that *who substitutions* indicate a main clause interpretation for the following reasons: In each target story, there was only one location (e.g., *the campsite*) at which the main character *successfully* told someone about the embedded clause event (e.g., *Annie's horse riding*). However, there was often another location where the main character tried to explain to someone but ultimately failed (e.g., *at the stable with her instructor*). It was never the case that the main character tried to explain to the *same person* at two different locations; thus, the identity of the person being told systematically varied along with the location. Thus, a reference to the particular person (or group of people) uniquely involved in the main clause event can only indicate a main clause interpretation of the *wh*-question. Under this interpretation, *who substitutions* can be interpreted as a type of location information, as in *Annie explained (at the place) with her friends*.

Anecdotal evidence also supports this interpretation. All adults who used *who substitutions* in filled-gap conditions also provided at least one sublocation response. When asked about their response strategy, they said that if a sublocation was not easily identified from the cartoon image then they resorted to describing *who* was there rather than being redundant and giving the main clause location. We believe this evidence supports a main clause interpretation of these *who* responses. In the following analyses, however, we keep the *where* main clause responses separate from the *who* main clause responses for ease of interpretation.

3.1.5. Results: Proportion of responses from Experiment 1

For all five experiments, we report the relative proportion of each response type (by age group, if relevant). We excluded all “other” responses, as there were only a few across all of our experiments (< 7% in any condition). The proportion of responses from Experiment 1 is summarized in Figure 2. In the globally ambiguous condition, both adults and children provided more main clause responses than any other response type (adults = 89.6%, *SE* = 2.9%; children = 56.2%, *SE* = 6.0%). Neither group provided any sublocation responses. Both adults and children provided *who substitutions* (adults = 8.3%, *SE* = 2.8%; children = 4.2%, *SE* = 1.4%) and embedded clause responses (adults = 2.1%, *SE* = 1.0%; children = 39.6%, *SE* = 6.2%). Interestingly, the children gave more embedded clause responses than the adults in the ambiguous condition. We return to this point later in the Discussion.

In the filled-gap condition, both adults and children provided all four response types: (incorrect) main clause locations (adults = 13.9%, *SE* = 4.8%; children = 25%, *SE* = 5.1%), grammatically permissible sublocations (adults = 24.3%, *SE* = 4.8%; children = 14.6%, *SE* = 3.6%), *who substitutions* (adults = 13.9%, *SE* = 4.0%; children = 33.3%, *SE* = 6.0%) and embedded clause responses (adults = 47.9%, *SE* = 7.3%; children = 27.1%, *SE* = 5.9%). In filled-gap conditions, children preferentially responded with *who substitutions*, whereas adults preferred embedded clause responses.

3.1.6. Statistical analyses

For our statistical analyses of Experiment 1, we recoded the data to explore two different binary dependent variables: (i) all of the *where* main clause responses, including sublocations but excluding *who substitutions*

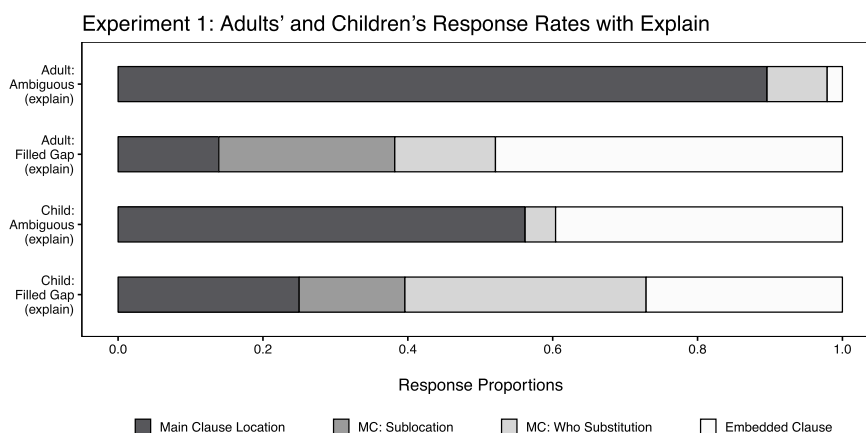


Figure 2. Average by-participant mean proportion of responses for both adults and children in Experiment 1. Responses consistent with main clause events are main clause location (black), sublocations (dark gray), and who substitution (gray) responses.

(1 = main clause response, 0 = other), and (ii) all of the embedded clause responses (1 = embedded clause response, 0 = other). We were unable to fit models for the *who* responses, thus for all five experiments, we discuss the qualitative patterns for these responses rather than providing statistical analyses.³ We analyzed both binary measures using generalized logistic mixed effects models with fixed effects of age group (adults vs. children) and *wh*-question type (ambiguous vs. filled-gap) as well as their interaction. We implemented our models with the *lme4* package in the R statistical computing environment (R Core Team 2018; Bates et al. 2014). We initially fit all models with maximal random effects structures (i.e., random slopes and intercepts for both subjects and items). Most maximal models did not converge; thus, we reduced each model until reaching model convergence (see Baayen, Davidson & Bates 2008; Barr et al. 2013; Jaeger 2008). In Experiment 1, the main clause model only had random intercepts for subjects and items. The embedded clause model had random intercepts for subjects and items, as well as random slopes for subjects. Both models used simple difference coding for the fixed effects: age group (adults = 0.5, children = -0.5) and *wh*-question type (ambiguous = 0.5, filled-gap = -0.5). We evaluate significance as a *z* value with absolute value greater than 2 as convention for $\alpha = .05$ (see Baayen 2008; Gelman & Hill 2007).

3.1.7. Results: Statistical analyses from Experiment 1

The output of both generalized logistic mixed effects models is summarized in Table 2. All significant effects are marked with asterisks. The main clause model only revealed a main effect of *wh*-question type ($\beta = 5.42$, SE = 2.25, $z = 2.41$, CI: [1.00, 9.84]) as more main clause responses occurred in ambiguous conditions compared to filled-gap conditions.⁴ Interestingly, there was no effect of age

³Across all experiments, *who* substitution responses had a particular distribution that resulted in complete/partial separability issues when trying to fit the data. Essentially, these responses could be (almost) perfectly predicted by a linear combination of our condition and subject variables. This separability led to unreliable model outputs with extreme standard errors and *p* values close to (if not exactly) 1. To circumvent these issues, we fit these data using generalized logistic mixed effects models within a Bayesian framework, which by-passes issues of separability with binary response variables (see Abrahantes & Aerts 2012; Kimball et al. 2018). We do not discuss these analyses in depth, but we provide footnote comments about the results of those Bayesian models throughout the article. Again, all statistical analyses can be found on our OSF (osf.io/qc84n/).

⁴We briefly acknowledge the potential issues of generalizability given our small sample sizes, between-participant design, and potential difference between items (see Experiment 5). To address any concerns, we fit all of our models within Frequentist and Bayesian frameworks. We then calculated the respective confidence and credible intervals for all model estimates and compared the range of the intervals. In all comparisons, the Frequentist intervals were wider than the Bayesian intervals, suggesting greater uncertainty in our Frequentist estimate. The most conservative approach would be to present the results from the Frequentist models, as they represent a greater degree of variability and continue to show significant patterns (see Kimball et al. 2018 for further discussion).

Table 2. Summary of model output for Experiment 1.

Main clause model: Effect	Estimate (SE)	z
Age group	2.91 (1.80)	1.61
Wh-question type*	5.42 (2.25)*	2.41*
Age group*wh-question type	4.82 (3.37)	1.43
Embedded clause model: Effect	Estimate (SE)	z
Age group	-0.72 (2.88)	-0.25
Wh-question type	-2.47 (2.71)	-0.91
Age group*Wh-question type*	-12.51 (5.60)*	-2.23*

group nor a significant interaction. This indicates that adults and children provided roughly similar (although descriptively different) proportions of main clause responses across conditions.

The embedded clause model only revealed a significant interaction between age group and *wh*-question type ($\beta = -12.51$, $SE = 5.60$, $z = -2.23$, $CI: [-23.48, -1.54]$). Planned pairwise comparisons by age group revealed a significant effect of *wh*-question type for adults ($\beta = 4.54$, $SE = 2.08$, $z = 2.18$) but not children ($\beta = -1.78$, $SE = 1.70$, $z = -1.05$). This indicates an increase in adults' embedded clause responses in the filled-gap condition, whereas children continued to provide the same amount of embedded clause responses in both ambiguous and filled-gap conditions.

Both adults and children provided *who* substitution responses in the ambiguous (adults = 8.3%, $SE = 2.8\%$; children = 4.2%, $SE = 1.4\%$) and filled-gap conditions (adults = 13.9%, $SE = 4.0\%$; children = 33.3%, $SE = 6.0\%$), suggesting similar availability of this response strategy across age groups.⁵

3.1.8. Discussion

Experiment 1 investigated the comprehension of ambiguous and garden-path *wh*-questions by English-speaking adults and 5-year-old children using a between-participants manipulation of *wh*-question type. Critically, this experiment used *wh*-questions that were directly modeled after the French versions used by Lassotta, Omaki & Franck (2016). In the following sections, we describe each response type individually, starting with main clause responses.

3.1.8.1. Main clause responses. We found that both English-speaking adults and children show main clause preferences in the ambiguous condition, replicating earlier findings in French (Lassotta, Omaki & Franck 2016), although our adults' preference for main clause responses was much greater (89.6%) than children's (56.2%) for ambiguous *wh*-questions. One possible interpretation of this difference is that adults' active gap-filling biases may be stronger than children's biases—but this interpretation remains speculative, given the lack of a significant interaction between age group and *wh*-question type in the main clause model.T 2018; ;).

In the filled-gap condition, both adults and children continued to provide (incorrect) main clause locations (adults = 13.9%; children = 25%) but to a lesser degree than in the ambiguous condition. We also observed the first use of sublocation responses (adults = 24.3%; children = 14.6%). Sublocation strategies are not always available to participants. For example, Japanese adults and children do not use this response strategy, as sublocations are argued to be pragmatically infelicitous given the *wh*-questions in (3). Omaki et al. (2014) explain that if a Japanese speaker intended to discuss a sublocation, it would be more natural to use *where* with a genitive marker (*-no* in Japanese) that directly selects the target locative PP (see Omaki et al. 2014 for further discussion). In French, however, these strategies are felicitous and often used by French-speaking adults in the filled-gap condition (18%). Lassotta, Omaki & Franck (2016) did not observe any sublocation responses from French-speaking children; however, we believe this is due to their methodology. French-speaking children provided their responses by touching a location on the screen. Simply touching a location

⁵Bayesian analyses of *who* responses in Experiment 1 revealed no main effects or interactions, suggesting no robust differences between adults' and children's use of this response strategy in both *wh*-question conditions.

does not clearly discriminate between a sublocation and the larger main clause location. Thus, we would expect French-speaking children to pursue this strategy, if required to give a verbal response like the French-speaking adults.

3.1.8.2. Embedded clause responses. English-speaking adults dispreferred embedded clause interpretations in ambiguous conditions (2.1%) but preferred them in filled-gap conditions (47.9%). English-speaking children, however, provided similar rates of embedded clause responses across ambiguous (39.6%) and filled-gap conditions (27.1%). Again, this pattern replicates the earlier findings from French: Adults mostly switch to embedded clause responses in the filled-gap conditions, whereas children continue to pursue some type of main clause response (Lassotta, Omaki & Franck 2016). We take this as evidence that English-speaking children (like French-speaking children) struggle to suppress their main clause bias and adopt the embedded clause interpretation. Interestingly, English-speaking children provided a higher amount of embedded clause responses (ambiguous = 39.6%; filled-gap = 27.1%) than French-speaking children (ambiguous = 15%; filled-gap = 12%). We take this increase in embedded clause responses as suggestive evidence that English-speaking children have a stronger ability to inhibit their initial biases in favor of an alternative interpretation, potentially resulting from their knowledge of obligatory *wh*-fronting in English. We return to this point in the General Discussion.

3.1.8.3. Who substitutions. The emergence of *who* substitutions is arguably the most striking difference between the present study and the previous cross-linguistic literature. *Who* substitutions are unattested in the previous French and Japanese studies, as well as the English study with ambiguous *wh*-questions (Lassotta, Omaki & Franck 2016; Omaki et al. 2014). However, the present study finds English-speaking adults and children describing *who* was at the main clause event (e.g., *her friend*) rather than *where* it took place. While we had not predicted this type of response, given the previous data, in retrospect they seem somewhat sensible given the pragmatics of our *wh*-questions in the context of our stories.

In Experiment 1, each target story followed the main character as they performed some theme-relevant task (e.g., *horse riding*) and searched for someone to tell about that task. More specifically, the main character tries to tell two different people (or groups of people) at two different locations. For example, Annie first tries to tell *her riding instructor at the stable* but fails. Later on, she tries to tell *her friend at the campsite* and succeeds. This story design sets up a salient contrast between two people: one that was told successfully and one that was not. This contrast may have caused participants to focus on the identity of the person at the *successful* main clause location. Moreover, the identity of each person varies with location, as Annie did not try to tell *the same person* at two different locations. Thus, any reference to a particular person is also an indirect reference to a particular location, i.e., *Annie explained (at the place) with her friend*. In a sense, *who* substitution works like the sublocation repair strategy, where listeners provide a narrower description, e.g., *by the tent (at the campsite)*, to preserve their main clause interpretation.⁶ If *who* substitutions act as a main clause repair strategy, this would also explain why adults and children primarily gave *who* substitutions in the filled-gap conditions (in ambiguous, adults = 8.3%; children = 4.2%; in filled-gap, adults = 13.9%; children = 33.3%).

The next four experiments further investigate the source of these *who* substitutions. Experiment 2 is a partial replication of the original English study by Omaki et al. (2014) that asked ambiguous *wh*-questions like (6a). Experiment 2 adopts their use of *tell someone* (6) and attempts to replicate their original findings, as well as our findings from Experiment 1.

- (6) a. Where did Annie tell someone that she rode her horse?
- b. Where did Annie tell someone **at the campsite** that she rode her horse?

⁶In fact, all adults who provided *who* substitutions in the filled-gap conditions for Experiment 1 also provided at least one sublocation response as well.

To preview our results, Experiment 2 replicates our conclusions from Experiment 1. We also replicated Omaki et al. (2014), including their lack of *who* substitutions in the ambiguous conditions. However, we found that children continued to use *who* substitutions to repair filled-gap *wh*-questions like (6b). Experiments 3–5 investigate why children continue to use *who* substitution strategies: Experiment 3 investigates presentation rate; Experiment 4 investigates the definiteness of the indirect object; and Experiment 5 is a partial replication of Experiment 4. Table 3 provides a summary outline of the experiments in the rest of the article.

3.2. Experiment 2: Ambiguous and filled-gap *wh*-questions using ‘tell someone’ constructions

Experiment 1 investigated how adults and children interpret both globally ambiguous and filled-gap, bi-clausal *wh*-questions using main clause verb *explain*. In Experiment 2, we use *wh*-questions with the construction *tell someone* (6). The ambiguous condition (6a) is identical in Omaki et al. (2014), so we expected to replicate their earlier findings that English speakers prefer main clause interpretations of globally ambiguous *wh*-questions. The garden-path condition in (6b) is simply the filled-gap version of (6a). Since Experiment 2 uses a different main clause construction than Experiment 1 and Lassotta, Omaki & Franck (2016), any comparison between response patterns should be considered indirect. We reiterate the difficulties with interpreting the attachment of the PP in the filled-gap condition (6b) given the intervening NP between the verb and the filler. However, we will argue that English speakers did not seem to interpret the PP as modifying the indirect object *someone*, i.e., *Where did Annie tell [someone at the campsite]_{NP} that she rode her horse?*, and often remarked that the experimenter “told them the answer in the question” (see Discussion).

3.2.1. Participants

We recruited 26 adult native English speakers from the community at the University of Washington, Seattle. Adult participants also participated in another experiment unrelated to the present study during their visit. Twenty-five children were also recruited for participation from the surrounding community of King County, WA. Of those participants, we excluded two adults and one child from further analyses for providing more than two incorrect or irrelevant answers. Adult participants ($N = 24$) ranged in age from 18–49 (mean age = 24.3) and were compensated at a 15 USD/hour rate. The children in the ambiguous ($N = 12$) and filled-gap ($N = 12$) conditions ranged from 5;00 to 5;09 (mean age = 5;03) and 5;02 to 6;02 (mean age = 5;09) respectively. These age ranges roughly correspond to those in Omaki et al. (2014).

Table 3. Summary of all five experiments including question durations.

Exp.	Group		Condition	Question duration (SE)
Exp1	Adult	$N = 12$	Ambiguous	5.91 (0.39)
		$N = 12$	Filled-gap	7.46 (0.26)
	Child	$N = 12$	Ambiguous	5.53 (0.27)
		$N = 12$	Filled-gap	7.32 (0.25)
Exp2	Adult	$N = 12$	Ambiguous	3.99 (0.17)
		$N = 12$	Filled-gap	6.56 (0.34)
	Child	$N = 12$	Ambiguous	4.22 (0.17)
		$N = 12$	Filled-gap	6.85 (0.36)
Exp3	Child	$N = 12$	Filled-gap, slow	8.36 (0.43)
Exp4	Child	$N = 12$	Filled-gap, slow, definite NP	8.66 (0.48)
Exp5	Child	$N = 12$	Filled-gap, slow, definite NP	8.24 (0.66)

3.2.2. Materials

Experiment 2 used the same story stimuli and question conditions from Experiment 1.⁷ However, instead of using *explain* as the main clause verb, we used the verb *tell* with the indefinite pronoun *someone* (6), as this was the construction previously used in Omaki et al. (2014).

3.2.3. Procedure

The procedure for Experiment 2 is identical to Experiment 1. The same experimenter read the ambiguous *wh*-questions with an average duration of 3.99 ($SE = 0.17$) for adults and 4.22 ($SE = 0.17$) seconds for children. In the filled-gap conditions, questions were read with an average duration of 6.56 ($SE = 0.34$) and 6.85 ($SE = 0.36$) seconds for adults and children respectively. Again, the durations for the filled-gap conditions are longer than the ambiguous conditions due to the overt PP. Interestingly, the average durations for all conditions in Experiment 2 are *shorter* than those in Experiment 1 (see Table 3), even though Experiment 2 had longer main clauses (e.g., *tell someone* vs. *explain*). Experiment 3 explores whether this slightly faster presentation rate in Experiment 2 influenced the pattern of responses for children.

3.2.4. Data coding and statistical analysis

The coding procedures and statistical analyses were identical to Experiment 1. We again created two binary response variables for the main clause and embedded clause responses. All generalized logistic mixed effects models were first fit with maximal random effects structures and then reduced until reaching model convergence. The main clause model had random intercepts for subjects and items and random slopes for subjects (excluding a slope for the interaction term). The embedded clause model only had random intercepts for subjects and items. Significant effects are indicated by an absolute value greater than 2. As in Experiment 1, we only discuss *who* substitutions descriptively.

3.2.5. Results

The response patterns by adults and children are summarized in Figure 3. In the ambiguous conditions, both adults and children overwhelmingly preferred main clause responses (adults = 79.2%, $SE = 4.8\%$; children = 87.5%, $SE = 2.9\%$) relative to embedded clause responses (adults = 20.8%,

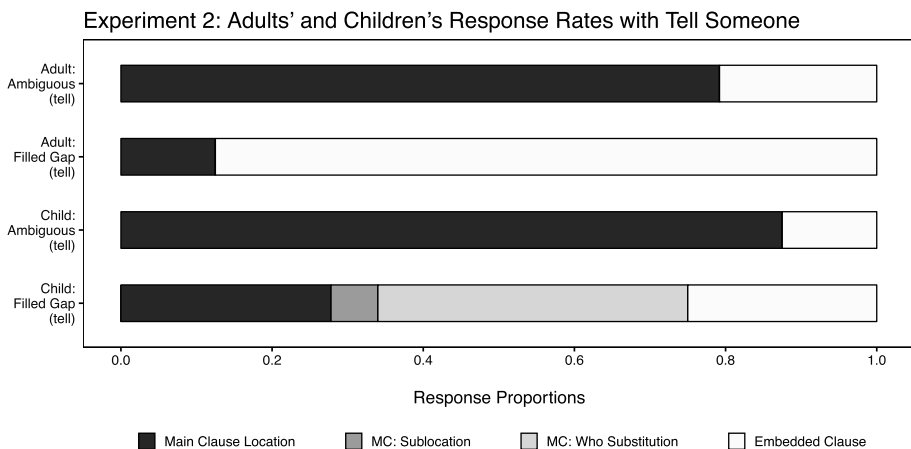


Figure 3. Average by-participant mean proportion of responses for both adults and children in Experiment 2. Responses consistent with main clause events are main clause location (black), sublocations (dark gray), and who substitution (gray) responses.

⁷Experiment 2 was actually conducted before Experiment 1. Throughout the five experiments, we realized that one target story seemed to overemphasize the embedded clause event (see Experiment 5 for more information). As such, Experiments 2–4 included that target story, and it was then replaced with a filler story in Experiments 1 and 5.

$SE = 4.8\%$; children = 12.5%, $SE = 2.9\%$). Compared to Experiment 1 (which used different constructions) children provided fewer embedded clause responses with 12.5% in Experiment 2 and 39.6% in Experiment 1. Neither group provided any sublocations or *who* substitutions in the ambiguous conditions. However, the lack of *who* substitutions in ambiguous conditions is not entirely unexpected given that they were relatively infrequent for ambiguous conditions in Experiment 1 (adults = 8.3%, $SE = 2.8\%$; children = 4.2%, $SE = 1.4\%$).

In the filled-gap condition, adults and children differed in their preferred responses: Adults strongly preferred embedded clause interpretations (adults = 87.5%, $SE = 4.2\%$; children = 25%, $SE = 3.9\%$), whereas children preferred *who* substitutions (adults = 0%; children = 41%, $SE = 6.0\%$) and other main clause interpretations (adults = 12.5%, $SE = 4.2\%$; children = 34%, $SE = 6.0\%$).

The output of the generalized logistic mixed effects models is summarized in Table 4. For main clause responses, there was only a main effect of *wh*-question type ($\beta = 6.59$, $SE = 2.51$, $z = 2.62$, CI: [1.66, 11.51]) with more main clause responses occurring for the ambiguous than the filled-gap conditions. There was no effect of age group nor a significant interaction. For embedded clause responses, there were main effects of *wh*-question type ($\beta = -3.36$, $SE = 0.97$, $z = -3.44$, CI: [-5.27, -1.45]) and age group ($\beta = 2.88$, $SE = 0.91$, $z = 3.17$, CI: [1.10, 4.67]). Both main effects were superseded by a significant interaction of age group and *wh*-question type ($\beta = -4.24$, $SE = 1.73$, $z = -2.45$, CI: [-7.63, -0.85]). Planned pairwise comparisons by age group revealed a significant effect of *wh*-question type for adults ($\beta = 3.05$, $SE = 0.83$, $z = 3.67$) but not children ($\beta = 0.58$, $SE = 0.63$, $z = 0.92$). Taken together, these results indicate that adults' main clause interpretation preference was restricted to the ambiguous *wh*-question, whereas children preferred main clause interpretations in both ambiguous and filled-gap conditions. Unlike Experiment 1, *who* substitutions were only provided by children in the filled-gap condition—and moreover, this child repair strategy was preferred over all other response types (41%, $SE = 6.0\%$).⁸

3.2.6. Discussion

3.2.6.1. Main clause and embedded clause responses.

Experiment 2 replicated the findings from Experiment 1 using a different main clause construction, i.e., *tell someone*, which was used previously by Omaki et al. (2014). Both English-speaking adults and children preferred main clause interpretations for ambiguous *wh*-questions (and to a greater degree than observed in Experiment 1). This preference constitutes further evidence for the shorter dependency bias, as previously observed in French and Japanese. In the filled-gap condition, we observed adults' preference for embedded clause responses and children's continued preference for some type of main clause interpretation (e.g., main clause locations, sublocations, *who* substitutions). This finding also replicates the results from Experiment 1 showing that English-speaking children struggle to suppress their main clause bias and adopt embedded clause responses. Interestingly, children in Experiment 2 provided fewer embedded clause responses and more strongly preferred main clause interpretations than in Experiment 1.

One explanation for fewer embedded clause responses in Experiment 2 could be the differences in main clause verbs. Omaki et al. (2014) report that children provide more embedded clause interpretations with

Table 4. Summary of model output for Experiment 2.

Main clause model: Effect	Estimate (SE)	z
Age group	-1.67 (2.41)	-0.69
<i>Wh</i>-question type*	6.59 (2.51)*	2.62*
Age Group* <i>Wh</i> -question type	1.68 (2.28)	0.74
Embedded clause model: Effect	Estimate (SE)	z
Age group*	2.88 (0.91)*	3.17*
<i>Wh</i>-question type*	-3.36 (0.97)*	-3.44*
Age group*<i>Wh</i>-question type*	-4.24 (1.73)*	-2.45*

⁸Bayesian analyses of *who* responses in Experiment 2 unsurprisingly revealed a significant interaction ($\beta = 6.45$, CI: [0.15, 13.81]) given that *who* substitutions only occurred in children's filled-gap conditions.

bare verbs, arguing that children may interpret the verb as a type of evidential marker (i.e., *According to Annie, where did she ride her horse?* For similar results with bare *say*, see de Villiers et al. 2008; de Villiers, Roeper & Vainikka 1990). However, the overall pattern of responses for English-speaking adults and children support the cross-linguistic evidence for shorter dependency biases in both adults and children.

3.2.6.2. *Who* substitutions. In Experiment 2, we saw a reduction in the amount of *who* substitutions for adults in both conditions and for children in the ambiguous condition. It is not entirely surprising that there were no *who* substitutions for the ambiguous *wh*-questions, given that they were relatively infrequent in Experiment 1 (8.3% for adults, 4.2% for children). Adults completely stopped using this repair strategy and relied more on the other main clause repair strategy, namely sublocations. One possible explanation for adults' lack of *who* substitutions is that by mentioning the indirect object in the question (i.e., *tell someone*) we made *who* responses somewhat redundant. In Experiment 1, adults reported using *who* substitutions i) to avoid saying the main clause location again (a redundant response) and ii) if a sublocation was not easily inferred from the cartoon image. This does not explain why children continued to use this repair strategy in the filled-gap conditions. However, the next three experiments were designed to further investigate why children continue to rely on *who* substitution strategies.

3.3. Experiment 3: Presentation rate manipulation for filled-gap *wh*-questions

In Experiment 1, English speakers frequently gave *who* substitution responses (e.g., “her friend”) to questions like *Where did Annie explain at the campsite that she rode her horse?* In Experiment 2, adults no longer provided these responses and children only used *who* substitution strategies in filled-gap conditions. One difference between Experiments 1 and 2 was the rate of question presentation (see Table 3). In Experiment 1, the researcher presented both question conditions more slowly than those in Experiment 2. Speech rate is an important experimental parameter that is not always systematically controlled in offline, story-based comprehension tasks. As such, the faster rate of presentation in Experiment 2 could have made the filled-gap conditions more challenging for children. Thus, we decided to partially replicate Experiment 2 at a slower speed with English-speaking children to see if that impacts children's use of *who* substitutions.

3.3.1. Participants

In Experiment 3, we recruited a different set of 13 native English-speaking children from the communities at the University of Washington, Seattle and King County, WA. We excluded one child from analyses using the same exclusion criteria from Experiments 1 and 2. The remaining 12 children used in our final analyses were aged 5;03 to 6;01 (mean age = 5;08).

3.3.2. Materials

Experiment 3 used the same stimuli from Experiments 1 and 2. In this experiment, we only asked the filled-gap *wh*-questions from Experiment 2, as our goal was to investigate why children continued to use this repair strategy in filled-gap conditions.

3.3.3. Procedure

The procedure for Experiment 3 follows exactly from the previous experiments. The only change was the speed of the question presentation. In Experiment 3, the filled-gap conditions were read more slowly, with an average duration of 8.36 seconds ($SE = 0.43$). For reference, the filled-gap conditions in Experiments 1 and 2 were read with average durations of 7.32 ($SE = 0.25$) and 6.85 ($SE = 0.36$) respectively. Remember that the questions in Experiment 2 have a longer main clause (*tell someone*), which is important when comparing their durations. We adopted this slower rate of presentation for Experiments 4 and 5 (see Table 3 for average question durations).

3.3.4. Data coding and analysis

The coding procedures were identical to the previous experiments. We compared the same two binary variables for main clause and embedded clause responses using generalized logistic mixed effects models. We had a fixed effect of question duration (fast vs. slow) with simple difference coding (fast = 0.5, slow = -0.5), which tested for differences between the response patterns in Experiment 2 and 3. The main clause model had random intercepts and slopes for both subjects and items. The embedded clause model had random intercepts for subjects and items and slopes for subjects only. Again, the proportions of *who* responses are only discussed descriptively.

3.3.5. Results

Response patterns in Experiment 3 are summarized in Figure 4, which also shows the data from children in the filled-gap condition in Experiment 2 to facilitate comparisons. The overall response patterns were very similar between Experiments 2 and 3. In the slowed, filled-gap condition (Experiment 3) children provided *who* substitutions at a slightly higher rate (52.8%, $SE = 13.5\%$) than in Experiment 2 (41%, $SE = 6.0\%$) and Experiment 1 (33.3%, $SE = 6.0\%$).⁹ In Experiment 3, children provided main clause responses (16.7%, $SE = 11.2\%$) and embedded clause responses (30.5%, $SE = 12.2\%$); however, there were no reliable changes from Experiment 2 with respect to children's main clause ($\beta = 6.82$, $SE = 8.45$, $z = 0.81$, $CI: [-9.74, 23.37]$) or embedded clause responses ($\beta = 0.10$, $SE = 1.57$, $z = 0.06$, $CI: [-2.98, 3.17]$).

3.3.6. Discussion

Experiment 3 examined whether children's continued use of *who* substitution responses in the filled-gap *wh*-questions was a product of the sentence presentation being too fast in Experiment 2. To address this, we used the same design as Experiment 2 while systematically slowing down the sentence presentation. However, we found no evidence for an effect of presentation rate, but we were able to successfully replicate the findings from Experiment 2 using the same materials.

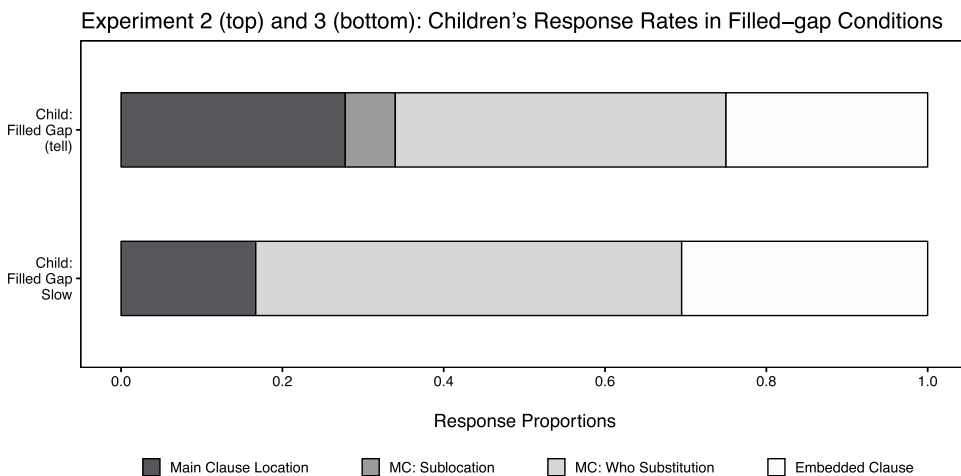


Figure 4. Average by-participant mean proportion of responses for children in two filled-gap experiments. Experiment 2 used a faster presentation rate for filled-gap *wh*-questions. Experiment 3 had the slower presentation rate. Responses consistent with main clause events are main clause location (black), sublocations (dark gray), and *who* substitution (gray) responses.

⁹Bayesian analyses directly comparing *who* responses in Experiment 3 (slowed, filled-gap) to those in Experiment 2 (filled-gap) revealed no significant difference between children's rate of *who* responses.

3.4. Experiment 4: Exploring the role of indefinite pronouns in *who* substitution errors

Experiments 1–3 showed that English-speaking 5-year-old children often make *who* substitution errors when answering garden-path *wh*-questions with and without indirect objects. Experiment 3 also showed that question presentation rate does not reliably affect the pattern of children responses. In Experiment 4, we investigate another potential source of children's continued use of *who* substitutions—the indefinite pronoun *someone*.

Our previous explanation for *who* substitutions was the following: Listeners are aware that the presence of PP in the main clause (e.g., *at the campsite*) is incompatible with the main clause association of *where*. In response, they *repair* their perception of the sentence input, specifically the *wh*-phrase, in such a way that they are still able to retain a main clause association. Given the pragmatics of our stories, listeners may have focused on the indirect object of the telling as a potential target for the *wh*-phrase. In other words, listeners reinterpret the main clause portion of the sentence as something that preserves their initial main clause interpretation like *Who was at the campsite where Annie successfully explained?*

This pragmatic account also explains the lack of *who* substitutions in French, as their stories did not set up a contrast between two possible recipients of the *explaining*. There was only ever one person being told something, which made the indirect object less likely to be targeted by the *wh*-phrase. This account, however, does not explain why English-speaking adults used *who* substitutions in Experiment 1 but not in Experiment 2. We argued that the explicit mention of the indefinite pronoun was enough to make adults feel like *who* answers were redundant and irrelevant. Children do not seem as sensitive to the infelicity of redundant responses, as most of their answers to filled-gap *wh*-questions are main clause locations. Children's insensitivity to redundant responses would account for why they continued to use this repair strategy in Experiment 2.

An alternative explanation is that children may not interpret the indefinite pronoun as redundant information about the recipient of the telling event. Moreover, children may see the indefinite pronoun as a cue that the indirect object of the main verb is still a plausible target for *wh*-questions. We argue that this is a plausible strategy for several reasons: First, indefinite pronouns like *someone* are often used in contexts in which the referent is unknown to the speaker (e.g., *Someone cleaned the room, but I don't know who did*). So, it is reasonable to think that the speaker may have intended to ask questions about its referent. In Japanese, indefinite pronouns can only be used when the speaker is uncertain about the referent. For example, *I'm going to Europe with someone next week* is infelicitous in Japanese, unless the speaker actually does not know who they are traveling with. In the context of the present experiment, it would indeed be infelicitous for the experimenter to use an indefinite pronoun as the indirect object of the verb *tell* because the experimenter should know who it was.

Second, typology research has shown that *wh*-phrases and indefinite pronouns have very close morphological and syntactic relations. For example, Japanese indefinite pronouns are systematically derived from *wh*-phrases by adding question particles (e.g., *dare* 'who,' *dare-ka* 'someone'; *nani* 'what,' *nani-ka* 'something'). In a survey of 100 languages sampled from a variety of language families, Haspelmath (1997) observed that 63 languages have Japanese-like interrogative-based indefinite pronouns, suggesting that the connection between *wh*-phrases and indefinite pronouns is cross-linguistically common (see also Bruening 2007; Cheng 1991). In fact, while English does not have a productive paradigm of interrogative-based indefinite pronouns, some indefinite adverbials do contain *wh*-phrases (e.g., *somewhere*, *somehow*). We do not intend to argue that children actually misperceived *someone* as the *wh*-phrase *who*, but the typologically attested link between indefinite pronouns and *wh*-phrases may have facilitated children's use of *someone* as a signal to reference the recipient of the *telling* (for evidence that typological distributions of syntactic structures affect children's syntactic development, see Culbertson & Newport 2015).

To test whether indefinites further prompted children's use of this sentence repair strategy, Experiment 3 replaced the indefinite pronoun *someone* with an appropriate definite NP for each story (7).

(7) Where did Annie tell *her friend* **at the campsite** that she rode her horse?

If the indefinite pronoun signaled to children that the recipient is still a potential target for the *wh*-phrase, we predict that in its absence, children should no longer make such errors in response to filled-gap *wh*-questions. A potential outcome of this stimulus change is that children resort to other main clause interpretation responses, including main clause location responses (e.g., *at the campsite*) and sublocation responses (e.g., *by the tent*). This is consistent with the shorter dependency biases that have been observed in French and Japanese, and such results would further strengthen the evidence for shorter dependency bias in English as well.

3.4.1. Participants

We recruited 13 different native English-speaking children from the communities at the University of Washington, Seattle and King County, WA. We excluded one child from analyses using the same exclusion criteria from the previous experiments. The remaining 12 children used for our analyses were aged 5;00 to 6;01 (mean age = 5;07).

3.4.2. Materials

We used the same materials from Experiments 2 and 3. Similar to Experiment 3, we only asked the filled-gap *wh*-questions, but we replaced the indefinite pronoun *someone* with a definite NP. The definite NPs included *the scientist*, *her sister*, *her friend*, and *her friends*, and we used pronominal genitives when they seemed more contextually appropriate than a definite article *the*.

3.4.3. Procedure

The procedure was identical to Experiment 3, including the slowed presentation rate. The same experimenter read the filled-gap condition with the explicit, definite NP with an average duration of 8.66 ($SE = .48$) seconds, which overlaps with the average of 8.36 ($SE = .43$) seconds from Experiment 3. Note that the slight increase in the average duration for Experiment 4 is attributed to the definite NPs (i.e., *the scientist*, *her friends*), which were typically longer in duration than *someone*.

3.4.4. Data coding and analysis

The response coding was identical to the previous experiments. In this experiment, we compared the data in Experiment 3 (the filled-gap, *wh*-question with indefinite pronouns) and Experiment 4 (the filled-gap *wh*-questions with definite NPs) to assess the influence of indefinite pronoun vs. definite NPs on children's response types. We again used two binary response variables for the main clause and embedded clause interpretations. We had a fixed effect of NP type (definite NP, *her friend* vs. indefinite pronoun, *someone*) that was coded using simple difference coding (definite NP = .5, indefinite pronoun = - 0.5). The main clause model had random intercepts and slopes for both subjects and items. The embedded clause model had random intercepts for subjects and items and slopes for subjects only. Significance was evaluated using the same conventions as the previous experiments.

3.4.5. Results

The data patterns for Experiments 3 and 4 are summarized in Figure 5. In Experiment 4, we observed no *who* substitution responses. The children provided an equal percentage of main clause (50%, $SE = 11.5\%$) and embedded clause (50%, $SE = 11.5\%$) responses to the filled-gap *wh*-questions with a definite NP. The main clause responses consisted of both (incorrect) main clause locations (37.5%, $SE = 11.7\%$) and sublocations (12.5%, $SE = 9.0\%$). In Experiment 3, children provided fewer main

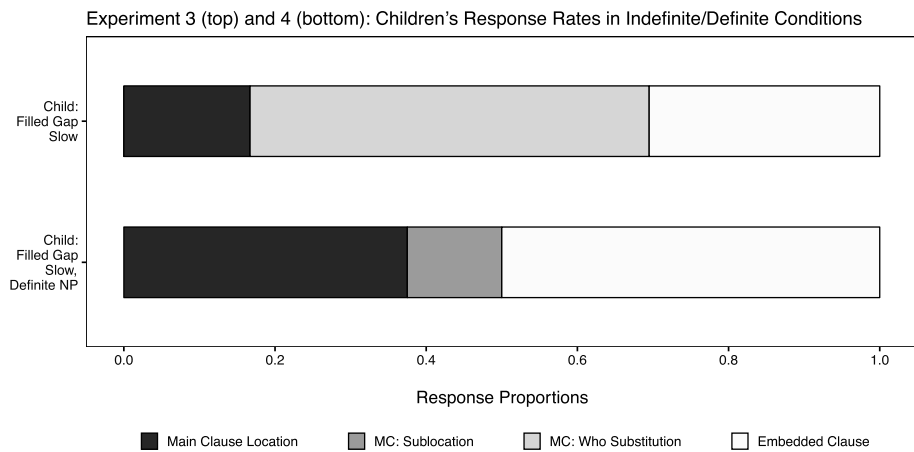


Figure 5. Average by-participant proportion of responses for children in two filled-gap experiments. Experiment 3 had slowed presentation of filled-gap *wh*-questions with indefinite pronouns. Experiment 4 had the slowed presentation rate but used definite NPs. Responses consistent with main clause events are main clause location (black), sublocations (dark gray), and who substitution (gray) responses.

clause responses (16.7%, $SE = 11.2\%$) and fewer embedded clause responses (30.6%, $SE = 12.2\%$) than in Experiment 4. While both response types increased with the elimination of the *who* substitutions, we only observed a significant change in the amount of main clause responses ($\beta = 6.07$, $SE = 2.92$, $z = 2.08$, $CI: [0.35, 11.78]$) and not in the amount of embedded clause responses ($\beta = 2.94$, $SE = 2.12$, $z = 1.39$, $CI: [-1.22, 7.10]$). This suggests that when *who* substitution responses are blocked, children's shorter dependency bias leads them to resort to other available *main clause* options.

One important caveat to Experiment 4 is that we observed for the first time a potential story bias for certain responses (which we briefly alluded to in the previous experiments). Visual inspection of the data suggested that one of the target stories (Item 1; *Jeff the astronaut*) tended to elicit more embedded clause responses on average (75%, $SE = 13.1\%$) than other stories, possibly because of the exciting nature of the embedded clause location or event (*Mars, where Jeff discovered aliens*). To investigate potential cross-item influences from this biased item, we explored the detailed distribution of embedded clause interpretations across items and lists. Table 5 presents by-item mean percentages of embedded clause responses per list.

This biased item (Item 1) was presented as the first target item in List 1 and last target item in List 2. Table 5 shows how this biased item affected responses in subsequent trials in List 1. The children in List 1 provided 19 embedded clause responses out of 24 data points, including 100% embedded clause responses for Item 1. In contrast, when Item 1 was the last trial (i.e., List 2 order), the children provided only 5 embedded responses out of 24 observations collectively. Moreover, three out of those five embedded clause responses came from Item 1, and no single child provided more than 2

Table 5. Summary of by-item embedded response rate for Experiment 4 by list presentation.

List number	Item in presentation order	Embedded response rate (SE)
List 1	1	100 (0)
	3	66.7 (14.9)
	5	83.3 (11.8)
	7	66.7 (14.9)
List 2	7	16.7 (11.8)
	5	0 (0)
	3	16.7 (11.8)
	1	50 (15.8)

embedded clause responses across trials. These results suggest that this particular biased item may have primed embedded clause responses in subsequent trials in List 1 and inflated the overall rate of embedded clause responses.

3.4.6. Discussion

In this experiment, we tested the hypothesis that the indefinite pronoun *someone* prompted children's continued use of *who* substitution strategies in Experiments 2 and 3. The question stimuli in this experiment replaced the indefinite pronoun with a definite NP, and this manipulation eliminated children's *who* substitution repair strategy. We saw that both main clause location (including sub-location responses) and embedded clause location response rates were 50%, which is a numerical increase for both response types compared to previous experiments (although only main clause responses significantly increased from Experiment 3).

This significant increase in main clause interpretation responses is consistent with previous observations that children struggle to inhibit their shorter dependency bias when processing filler-gap dependencies. Thus far, children have been relatively consistent in their use of embedded clause responses. However, the slight increase of embedded clause responses in Experiment 4 may have been due to the novel story bias from one particular item. It is interesting to note that such story biases were not observed in Experiments 2 and 3.¹⁰ So it is plausible that the unavailability of the *who* substitution responses in this experiment caused children to resort to an alternative source of information (e.g., event prominence) to arrive at another felicitous response.

In addition, Item 1 is the only target story in Experiments 2–4 that does not have the main character trying to tell two different people about the embedded clause event (*his trip to Mars*). Recall that Annie tries to tell her riding instructor about her ride but fails, so she goes and tells her friend at the campsite. In this story, Jeff only tells the scientist about his trip before he leaves. As such, this item does not provide a rigorous test of how the use of indefinites impacts children's reliance on *who* substitution strategies. In Experiment 5, we replace this biased item with one that uses the two-person story design and better equates the salience of the main clause and embedded clause events.

3.5. Experiment 5: Partial replication of Experiment 4 with a new story

The results from Experiment 4 showed that replacing *someone* with a definite NP eliminated children's *who* substitution responses for filled-gap *wh*-questions. However, further analyses of child response data revealed a specific item effect where children systematically perseverated on embedded clause responses after a particular item. In Experiment 5, we provide a more rigorous test of our indefinite hypothesis by replacing this biased target story with a story that (i) has a more balanced event and location prominence, (ii) includes two potential recipients of the *telling*, and (iii) has been previously used as a filler story in Experiments 2–4. This change allowed us to better assess when English-speaking 5-year-old children use *who* substitutions and whether children continue to provide as many embedded clause responses to filled-gap *wh*-questions without an influence from a biasing story.

3.5.1. Participants

We recruited 13 different native English-speaking children from the communities at the University of Washington, Seattle and King County, WA. We excluded one child based on the same criteria used in the previous experiments, resulting in 12 children aged 4;11 to 5;10 (mean age = 5;05).

3.5.2. Materials

We used the same task and stories from Experiments 2–4. We swapped the biased target story (Item 1: *Jeff the astronaut*) with a filler story (Item 2: *Bill playing soccer*) and adapted the target and filler

¹⁰The biased item from Experiments 2–4 was not a target item in Experiment 1. This is because Experiment 1 was conducted after Experiments 2–5 but was presented first for rhetorical purposes.

questions to match (see the Appendix for the entire list of these items and questions, including the changes in these two items). The filler and target set used in Experiment 5 is identical to Experiment 1.

3.5.3. Procedure

The procedure was identical to Experiment 4. A new experimenter, trained on question presentation, read the filled-gap condition with the definite NP with an average duration of 8.24 ($SE = 0.66$) seconds, which overlaps with the average of 8.66 ($SE = 0.48$) seconds from Experiment 3. The slight decrease in average duration for Experiment 5 is partly attributed to the change in target items.

3.5.4. Data coding and analysis

The coding and statistical procedures were identical to those in Experiment 4. We first tested whether the response patterns from Experiments 4 and 5 were significantly affected due to the biased item. To preview our results, there were no significant effects in any model comparing Experiments 4 and 5. We then compared Experiment 3 (slowed, filled-gap with indefinite pronoun) and Experiment 5 (slowed, new item, filled-gap with definite NP). We had a fixed effect of NP type with simple difference coding (definite NP = .5, indefinite pronoun = -0.5). The main clause model only had random intercepts for subjects, as no other model converged. The embedded clause model had random intercepts for both subjects and items.

3.5.5. Results

The comparisons of Experiments 4 and 5 and then Experiments 3 and 5 are summarized in Figures 6 and 7 respectively. For by-item embedded clause response rates in Experiment 5, see Table 6. We replicated the absence of *who* substitution responses (0%) from Experiment 4. Children's main clause response rates were 56.2% ($SE = 11.6\%$) and embedded clause responses were 43.8% ($SE = 11.6\%$). The main clause interpretation responses consisted of both main clause location responses (31.2%, $SE = 9.3\%$) and sublocations (25%, $SE = 9.7\%$). Moreover, as shown in Table 6, we no longer see a clear difference between List 1 (embedded clause = 45.83%, $SE = 4.18\%$) and List 2 (embedded clause = 37.5%, $SE = 10.5\%$) or evidence that early trials influenced later trials. This shows that changing the experimental items had the desired effect of removing the influence of a biased story.

The main clause model revealed a main effect of NP type ($\beta = 6.46$, $SE = 2.73$, $z = 2.37$, CI: [1.11, 11.81]) such that children provided more main clause responses in the definite NP (Experiment 5) condition than in the indefinite pronoun (Experiment 3) condition. This finding is consistent with

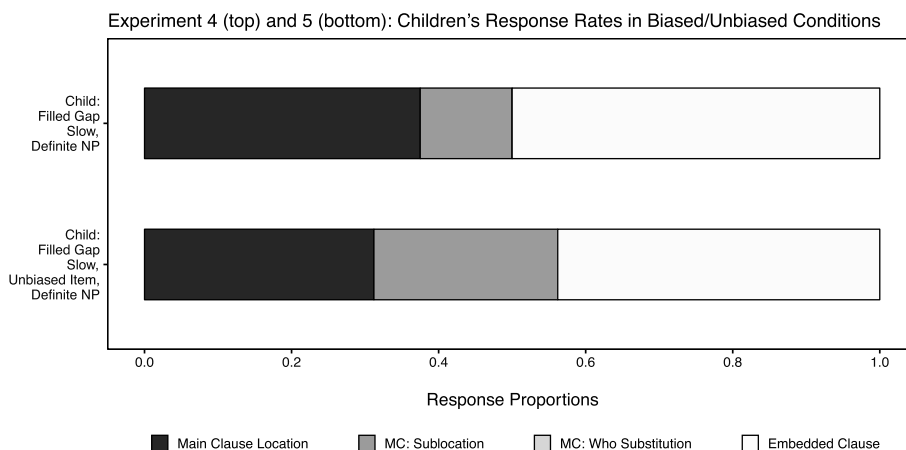


Figure 6. Average by-participant proportion of responses for children in two filled-gap experiments. Experiment 4 had the biased Item 1. Experiment 5 was the replication of Experiment 4 without the biased item. Responses consistent with main clause events are main clause location (black), sublocations (dark gray), and who substitution (gray) responses.

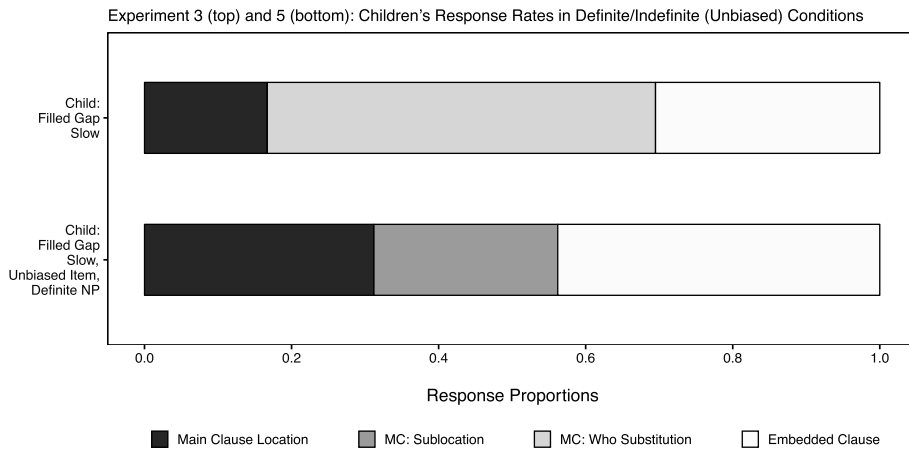


Figure 7. Average by-participant proportion of responses for children in two filled-gap experiments. Experiment 3 had the slowed presentation rate and indefinite pronouns. Experiment 5 was the replication of Experiment 4 without the biased item. Responses consistent with main clause events are main clause location (black), sublocations (dark gray), and who substitution (gray) responses.

Table 6. Summary of by-item embedded response rate for Experiment 5.

List number	Items in presentation order	Embedded responses (<i>SE</i>)
List 1	9 (unbiased)	50 (15.8)
	3	50 (15.8)
	5	50 (15.8)
	7	33.3 (14.9)
List 2	7	33.3 (14.9)
	5	33.3 (14.9)
	3	66.7 (14.9)
	9 (unbiased)	16.7 (11.8)

Experiment 4, reconfirming that replacement of the indefinite pronoun *someone* eliminated children's *who* substitutions, which in turn increased the proportion of other main clause responses. We also observed a slight decrease in embedded clause response rates from Experiment 4 (50.0%) to Experiment 5 (43.8%); however, the model comparing embedded clause responses showed no significant difference in the response rates across these two experiments ($\beta = -0.29$, $SE = 1.15$, $z = -0.25$, $CI: [-2.55, 1.97]$).

3.5.6. Discussion

Experiment 5 rigorously tested the hypothesis that the indefinite pronoun *someone* prompted children's continued use of *who* substitutions in Experiments 2 and 3. The improved stimuli in Experiment 5 successfully removed the item bias observed in Experiment 4 and demonstrated (again) that the filled-gap *wh*-question with a definite NP in the indirect object position blocks children's *who* substitution strategies.

Without the use of *who* substitution strategies, our English-speaking children primarily provided responses consistent with a shorter dependency bias (i.e., main clause interpretation answers, 56.2%). However, unlike the previous cross-linguistic studies, we found English-speaking children adopted a high amount of embedded clause responses (43.8%). In fact, this rate of embedded clause responses was largely consistent across all five experiments in the filled-gap conditions, even when children provided *who* substitutions (Exp1 = 27.1%; Exp2 = 25%; Exp3 = 30.6%; Exp4 = 50%; Exp5 = 43.8%). Children's embedded clause rates were certainly not as high as those of English-speaking adults (87.5%, as seen in Experiment 2), but their ability to entertain the second verb association appears

to be stronger than has been previously reported in French and Japanese-speaking children. We will explore potential reasons for this contrast in the following General Discussion.

4. General discussion

In five experiments, we investigated English-speaking 5-year-old children's processing of globally ambiguous *wh*-questions (8a) and their filled-gap counterpart (8b). Recent evidence in French and Japanese has shown that children overwhelmingly pursue first verb interpretations (e.g., *She explained at the campsite*) of *wh*-questions like (8).

(8) a. Where did Annie explain/tell someone that she rode her horse?

b. Where did Annie explain/tell someone **at the campsite** that she rode her horse?

Additionally, the cross-linguistic data show that children's preference to associate *wh*-questions with the first verb holds even in contexts where first verb locations (e.g., *at the campsite*) are not acceptable answers (8b). Given the prior findings, we asked whether English-speaking children show the same cross-linguistic preference for first verb associations in both ambiguous and filled-gap conditions. By looking at English, we were also able to fill a gap in the literature on children's comprehension of garden-path *wh*-questions in a head-initial language that *clearly* allows for long-distance *wh*-questions with cross-clausal filler-gap dependencies.

In Experiment 1, we show that English-speaking children indeed prefer main clause (first verb) interpretations of both globally ambiguous and filled-gap *wh*-questions; although English-speaking children provided a higher rate of embedded clause (second verb) responses than previously reported in the literature. Experiment 1 also documented the unique *who* substitution strategy, which was used primarily to repair garden-path *wh*-questions. We argued that the pragmatics of our stories made answers about *who* was at the main clause event rather than *where* it occurred somewhat felicitous. Each story set up a strong contrast between two potential recipients of the *explaining/telling* event, which may have caused participants to focus on the person who was successfully told. *Who* substitutions, like other main clause repair responses (e.g., sublocations), allowed participants to preserve their main clause bias in filled-gap conditions.

Experiment 2, however, found that adults no longer pursue *who* substitutions when an indirect object is explicitly mentioned (i.e., the indefinite pronoun in *tell someone*). One explanation is that adults find *who* answers redundant and uninformative once an indirect object is mentioned in the question, whereas children remain insensitive to the infelicity of redundant answers. This is consistent with the fact that most of children's responses were main clause locations, which were overtly mentioned in the question. An alternative explanation for children's continued use of *who* substitutions is that children do not interpret the indefinite pronoun as sufficient mention of the indirect object. Rather, children interpret *someone* as a cue that the speaker is asking about the person being told. Experiments 3–5 investigated this hypothesis, as well as others about the potential effects of presentation rate and item-biases. In these three experiments, we found that only changing the indirect object to a referential NP influenced children's use of *who* substitutions, providing support for the claim that children (unlike adults) interpret *someone* as a cue to continuing answering with the indirect object (e.g., *her friend*).

Taken together, the present findings provide convergent evidence for children's shorter dependency bias, which has been observed in both French and Japanese. Moreover, this bias persists in garden-path *wh*-questions where first verb associations are not grammatically permissible. The present study goes beyond the previous literature, however, by showing that English-speaking children seem to adopt more embedded clause interpretations than young speakers of French and Japanese. The next sections will explore the broader theoretical implications of these findings for developmental psycholinguistics and adult sentence processing research.

4.1. Shorter dependency biases as active gap-filling

Our findings show that children often associated the *wh*-phrase with the first verb in the sentence, and so far we have characterized this preference as a shorter dependency bias in filler-gap dependency processing. However, we have generally avoided calling this bias in children an active gap-filling bias, which is a more widely used terminology in the broader psycholinguistics community.

Part of the implicit reason for this terminological choice was that evidence for active gap-filling bias typically comes from *online* data, i.e., time course measures that demonstrate the incremental integration of the filler at the earliest, grammatically permissible position. These types of evidence have come from a wide variety of methodologies, including reading time contrasts in self-paced reading and eye-tracking-while-reading tasks (e.g., Aoshima, Phillips & Weinberg 2004; Omaki & Schulz 2011; Stowe 1986; Wagers & Phillips 2014; Omaki et al. 2015; Staub 2007; Traxler & Pickering 1996), incremental eye movement data during visual world eye-tracking (Atkinson et al. 2018; Sussman & Sedivy 2003), as well as N400 or P600 responses in ERP experiments (e.g., Chow et al. 2015; Garnsey, Tanenhaus & Chapman 1989; Kaan et al. 2000; Phillips, Kazanina & Abada 2005).

On the other hand, the present study, as well as the previous works in Japanese and French that laid the foundation for the present study (Lassotta, Omaki & Franck 2016; Omaki et al. 2014), all relied on *offline* responses alone with no time course measures of filler-gap dependency processing. As such, our measures do not necessarily provide evidence that the shorter dependency bias is a direct reflection of the online, active gap-filling bias during the initial processing of the filler-gap dependency; it is theoretically possible that children used a different bias in their initial processing, but somehow changed the initial interpretation in such a way that the ultimate interpretation happened to correspond to the interpretation outcome that is predicted by the active gap-filling bias.

We acknowledge the limitation of the present findings, but we argue on conceptual and empirical grounds that it is most reasonable to consider the present findings as evidence for the presence of an incremental, active gap-filling processing bias in children. First, as reviewed in the Introduction, perhaps the most robust finding in the last two decades of child sentence processing research is that children are extremely resistant to changing their initial, incrementally assigned interpretations (for reviews, see Omaki & Lidz 2015; Snedeker & Huang 2015). The skeptical view described above goes directly against this observation, as it presupposes that children are willing to subsequently change their initial analysis of filler-gap dependencies to match the active gap-filling bias.

Second, taken together with previous works in French and Japanese, we now have converging evidence across three languages that establishes the preference for first verb associations in globally ambiguous *wh*-questions and their garden-path versions. In Table 7, we summarize the offline interpretation preferences for both adult and child speakers of English, French, and Japanese. These

Table 7. First verb response rates for *wh*-questions in English, French, and Japanese.

Language	Condition	Age group	First verb response rate ^a	Preference
English	Ambiguous	Adult	90% (Exp1) 79% (Exp2)	Main clause
		Child	56% (Exp1) 88% (Exp2)	Main clause
	Filled-gap	Adult	38% (Exp1) 13% (Exp2)	Embedded
		Child	40% (Exp1) 56% (Exp5)	Main clause
French	Ambiguous	Adult	88%	Main clause
		Child	85%	Main clause
	Filled-gap	Adult	67%	Main clause
		Child	88%	Main clause
Japanese	Ambiguous	Adult	92%	Embedded
		Child	94%	Embedded
	Filled-gap	Adult	5%	Main clause
		Child	83%	Embedded

^aOmaki et al. (2014) and Lassotta, Omaki & Franck (2016) report all response rates rounded to whole proportions.
Note. All first verb responses include sublocation responses as well. We round our rates accordingly to make comparison easier. Also, the constructions from English Experiment 1 are closest to the French data, and Experiments 2 and 5 are closest to the Japanese data.

data reaffirm that the first verb associations (i) are preferred by adults when grammatically permissible (i.e., in ambiguous conditions), and (ii) are more generally preferred by children regardless of grammaticality or felicity. We attribute the persistence of children's first verb preference to their reluctance to abandon or suppress initial syntactic analyses. Moreover, children consistently demonstrate the same interpretation preferences in both ambiguous and garden-path *wh*-questions, which suggests that these first verb associations were most likely formed before the filled-gap cue (e.g., *at the campsite*) was presented in the input. Anecdotally, some children even explicitly said that the experimenter "told them the answer" when asking the question.

In sum, we argue that the developmental, cross-linguistic offline data from ambiguous and garden-path *wh*-questions altogether provide reasonable grounds for analyzing the first verb association bias in these offline experiments as a direct reflection of an incremental, active gap-filling bias. While the final confirmation of this conclusion awaits a version of online experiments with time course measures that maintain the same complexity of our target sentences and story design, we believe this tentative conclusion has useful methodological implications for future developmental research. As such, instead of simply dismissing offline data as suboptimal evidence for investigating incremental sentence processing, we would rather invite future research to explore the fine connections between time course measures and offline responses more—as a better understanding of the relation between incremental analyses and ultimate interpretations would allow us to construct better psycholinguistic models of how our interpretations change as a sentence unfolds for adults and children.

4.2. *Who* substitutions as a main clause repair strategy

We have argued that the processing mechanisms deployed by adults and children are informative with respect to understanding how interpretations are first constructed and ultimately obtained. We observed cross-linguistic similarities in adults' and children's processing bias to maintain first verb associations. For example, both English and French speakers entertain sublocation strategies as viable main clause interpretations of filled-gap *wh*-questions. While similar in their overall approach in parsing filled-gap *wh*-questions, only English speakers demonstrated a unique repair strategy (i.e., *who* substitutions) that has not been observed in the other cross-linguistic studies. Although unattested, *who* substitutions are similar to other repair strategies that allow for first verb interpretations of garden-path *wh*-questions. It is important to note, however, that this unique repair strategy most likely emerged due to methodological reasons rather than language-dependent reasons. In this remainder of this section, we provide a mechanistic description of how this processing procedure unfolds for English speakers comprehending the questions in our study (e.g., *Where did Annie explain **at the campsite** that she rode her horse?*)

First, listeners encounter the *wh*-phrase *where*, which prompts their active search for a gap position. In this question, the first gap occurs after *explain*. At this point, listeners posit the filler at the illusorily permissible position and then receive an error signal that their initial analysis is incompatible with the input. Second, since the presence of the PP blocks their first verb association, listeners are forced to make one of two processing choices: They could retain their main clause interpretation of *at the campsite* or they could simply abandon the main clause association and adopt an embedded clause interpretation. We show that English speakers ultimately select the former option to maintain the first verb association rather than reevaluate the association itself. Finally, listeners are now aware that their first verb association is incompatible with the current input; however, they are reluctant to give up this interpretation despite the conflicting information, presumably due to their overall bias to maintain their initial syntactic analyses. As a result, listeners select alternative processes that allow for a more felicitous main clause interpretation like sublocations or *who* substitutions.

We originally proposed that *who* substitutions are entertained for the following reasons: First, providing a description of *who* was present during the main clause event is informative with respect to *where* the event took place (e.g., *Annie explained (at the place) **with her friend***). In *wh*-questions that

omit explicit reference of the indirect object (i.e., *explain* in Experiment 1), *who* substitution strategies are available to both adults and children. However, in *wh*-questions that specify an indirect object using an indefinite pronoun (i.e., *tell someone* in Experiments 2–3), only children continue to use this repair strategy.

For adults, the presence of the indefinite pronoun may be sufficient information to make a *who* response redundant and no longer informative. Thus, instead of using *who* substitution, adults resort to the other main clause repair strategy, namely sublocations. For children, the indefinite pronoun may not sufficiently mention the indirect object to stop them from providing a redundant *who* response. Moreover, children seem insensitive to infelicitous, redundant responses, as evidenced by their large amount of main clause locations (e.g., *at the campsite*). Thus, it may be the case that for children having an implicit indirect object (*explain* in Experiment 1) is the same as having an underspecified, indefinite (*tell someone* in Experiment 2), both of which prompt *who* substitution responses.

Alternatively, children may interpret *someone* as a cue that the speaker is targeting the indirect object with the *wh*-phrase. We argued that this is a plausible explanation given that indefinite pronouns like *someone* are typically reserved for contexts with an unknown referent (e.g., *Someone cleaned the room, but I don't know who did it*). When a speaker uses *someone* in a *wh*-question, there is a possible interpretation that the target of the *wh*-question is actually about the referent of that indefinite in the main clause. The present study finds that children pursue *who* substitutions whenever the indirect object is not explicitly defined. This is consistent with the research on children's sensitivity to pragmatic and discourse factors related to referential expressions, which shows that children recognize changes in their interlocutor's choice of referential expressions (Clancy 1993, 1997; Matthews, Lieven & Tomasello, 2010; Graham, Sedivy & Khu 2014). As a result, children may find it infelicitous to not explicitly mention *who* after the narrator clearly labeled the person and it is obvious who was told something in the story. Our findings at least suggest a correlation between the presence of main clause indefinites and children's continued pursuit of *who* substitution strategies.

In sum, the observation of this unique sentence repair strategy calls for future research that investigates the details of what type of repair strategies are accommodated by adult and child comprehension mechanisms. It is well known in the adult psycholinguistics literature that even adults use sentence repair strategies when they struggle to assign a reasonable syntactic analysis to the input. For example, the famous garden-path sentence *The horse raced past the barn fell* is often misinterpreted to mean *the horse raced past the barn AND THEN fell*—as if the comprehender decided that they must have missed perceiving information in the input that would have made their initial parse of the sentence grammatical (Fodor & Inoue 1994). Similarly, we see material insertion in the comprehension of plausible intransitive sentences, especially those with unlikely agent/patient roles like “The rock kicked Mary” where word insertion would turn the intransitive sentence into a passive sentence “The rock WAS kicked BY Mary” (Gibson, Bergen & Piantadosi 2013; Levy 2008). The resulting passive sentence may be structurally complex, but the ultimate interpretation is more reasonable in a communicative situation.

Children are known to make similar *insertion* repair strategies; in the PP attachment ambiguity examples like *Put the frog on the napkin in the box*, children often made so called *hopping* errors, where they move the frog to the empty napkin and then again into the box (e.g., Trueswell et al. 1999). This hopping error could be taken to mean that the children repaired the sentence by insertion repair (e.g., *Put the frog on the napkin AND THEN into the box*) to make their initial PP attachment decision viable. Similarly, the famous interpretation errors involved in passive sentences can be considered as a case of deletion and substitution repair: When children misinterpret *The boy is gently kicked by the horse* as *The boy is gently kicking the horse*, they are engaging in the deletion of *is* as well as the substitution of *kicking* for *kicked by* (for recent eye-tracking evidence in children, see Huang, Leech & Rowe 2017; Huang et al. 2013).

Currently, there is much debate in the adult psycholinguistics literature on what type of repair strategies (*insertion, substitution, deletion*) are available, what degrees of repair (e.g., edit distance) are accommodated by the adult parser, and to what extent the repair decision is driven by the pressure to maintain the initial parse as opposed to the plausibility of the ultimate analysis (Gibson, Bergen & Piantadosi 2013). Given our observation of novel sentence repair strategies, we suggest that further investigations of what constraints guide both adults' and children's sentence repair behaviors, as well as how these constraints change through the course of development, could provide novel insights on these central questions in adult psycholinguistics.

4.3. The availability of embedded clause interpretations and grammatical biases

As we discussed in the Introduction, we suggested a possible reason why English-speaking children may entertain more embedded clause interpretations of garden-path *wh*-questions than French-speaking children. Our speculation was based on the somewhat surprisingly low rate of embedded clause responses to garden-path *wh*-questions in French-speaking adults: They only provided embedded clause responses 33% of the time. This rate was reliably higher than that of French-speaking children (12%) but much lower compared to the embedded clause response rate in English-speaking adults (87.5% in our Experiment 2).

The suggestion in Lassotta, Omaki & Franck (2016) was that this reflects grammatical differences in *wh*-question formation between French and English. English is an obligatorily *wh*-fronting language regardless of the gap position, whereas for long-distance bi-clausal questions, French allows both *wh*-fronting and in-situ *wh*-phrases, the latter of which is shown to be preferred in their acceptability judgment data. Lassotta, Omaki & Franck (2016) argued that this grammatical bias may have led French-speaking adults to primarily consider the fronted *wh*-phrase to have originated in the main clause, which accounts for why French-speaking adults struggle to associate the fronted *wh*-phrase with the embedded clause verb. Now, if this French-specific grammatical bias also affected French-speaking children's interpretation pattern, it is possible that children's main clause bias was somewhat inflated due to this developing grammatical knowledge of French *wh*-questions. On the other hand, if the English-specific knowledge of obligatory *wh*-fronting is emerging in 5-year-old children, they may also demonstrate an ability to adopt the embedded clause interpretation in garden-path *wh*-questions, despite the robust active gap-filling bias.

Our study sheds light on this question, especially in Experiment 1, which was maximally similar to Lassotta, Omaki & Franck (2016). We observed that English-speaking children provided embedded clause responses to garden-path *wh*-questions at a relatively high rate (27.1% in Experiment 1; 50% in Experiment 4; 43.8% in Experiment 5), unlike French-speaking 5- to 8-year-old children who only provided embedded clause responses 12% of the time. These descriptive contrasts between English and French are consistent with the hypothesis that developing language-specific knowledge of *wh*-question formation influences the development of children's ability to inhibit their active gap-filling bias.

Another open question is the difference between Japanese and English. While we have most directly compared French and English due to the word order similarity, we note that Japanese-speaking children's performance was closer to that of the French-speaking children rather than that of the English-speaking children (i.e., Japanese second verb association rate in the garden-path *wh*-question was low at 17%). One possible explanation is that perhaps the first verb interpretive commitments made in the embedded clause (structural position of the first verb in Japanese) are much harder to undo than similar commitments made in the main clause (structural position of first verb in English and French). From a pure structure building perspective, the main clause is always constructed *before* the embedded clause, regardless of the word order.

In English, due to its head-initial word order, the main clause input is entirely presented before any of the embedded clause input is encountered. As such, none of the embedded clause content is established when conflicts in the syntactic analysis occur at the main clause. This may make it relatively easier for English speakers to simply "pass the burden" onto the embedded clause, which

has yet to be structurally defined. In Japanese, on the other hand, the input and structural ordering is actually mixed: For example, the initial subject of the sentence always introduces the main clause structure, which includes a position for an (albeit empty) main clause verb. Although the main clause verb input is not encountered until the end of the sentence, its presence is strongly predicted by the presence of the main clause subject. As such, a structural “skeleton” for the main clause is already present early on in comprehension. The next step in processing involves the embedded clause input, which is where and when we believe the active gap-filling bias associates the filler with the embedded clause verb. At this point, even if the embedded verb association creates syntactic conflicts, any change to the upcoming main clause structure would involve altering the previously predicted (and arguably constructed) main clause structure. If we suppose that altering this predicted main clause structure is costly for Japanese children, we may be able to account for why Japanese children robustly avoid second verb associations in the presence of error signals.

These conclusions have implications for the future direction of research on sentence revision difficulties in children. Much of the work investigating the cause of children’s sentence revision difficulties has focused on the role of domain general cognitive mechanisms such as cognitive control (Novick, Trueswell & Thompson-Schill 2005; Qi, Fisher & Brown-Schmidt 2011; Woodard, Pozzan & Trueswell 2016). These studies have undoubtedly made important contributions to understanding how inhibitory control abilities outside of language may predict children’s inhibitory control process involved in garden-path sentence revision. However, the present study suggests that a better understanding of the development of language-specific constraints is also important in understanding how language-specific sentence revision mechanisms undergo development. These perspectives are obviously not mutually exclusive, and we invite future research that investigates how these cognitive and linguistic constraints interact in children’s sentence revision processes.

To conclude, we suggest that our findings support our original hypothesis that English-speakers more readily adopt embedded clause interpretations relative to speakers of other nonobligatory *wh*-fronting languages. These novel results provide further insight into the strength of the shorter dependency bias, as well as how the strength of that bias is influenced by language-specific constraints and syntactic knowledge on properties of *wh*-questions.

5. Conclusion

In five experiments, we show that English-speaking 5-year-old children’s processing of ambiguous and garden-path *wh*-questions can shed light on how general parsing biases toward shorter syntactic dependencies, as well as developing knowledge of language-specific grammatical constraints, impact the development of children’s filler-gap dependency processing mechanisms. We provide converging evidence that English-speaking children prefer main clause interpretations in both ambiguous and garden-path *wh*-questions, just like young speakers of French and Japanese. The present study also documents a unique sentence repair strategy to preserve their shorter dependency bias by addressing the indirect object of the main clause. Taken together, these findings support the evidence of cross-linguistic active gap-filling biases in adults and children, although we found that English-speaking children did pursue fewer main clause interpretations relative to the French and Japanese-speaking children in previous work, suggesting that speakers of obligatory *wh*-fronting languages like English may be more successful at abandoning first verb associations than speakers with optional *wh*-fronting as in French or Japanese.

In sum, we hope these findings inspire future research that investigates how language-specific constraints on *wh*-question formation affects filler-gap dependency processing in a wider variety of languages with different *wh*-question formation constraints, as well as how the interaction of language-specific constraints and broader cognitive constraints shapes the development of sentence revision mechanisms.

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Appendix

Following is the list of stories and questions used in Experiments 1 through 5. Experiments 1 use *explain*, Experiments 2–5 used the verb *tell*. Experiments 1 and 2 had both ambiguous and filled-gap conditions. Experiments 4 and 5 used the filled-gap questions with the explicit definite NP (e.g., *her friend*, *his mom*).

Target Stories:

Item 1

Jeff is an astronaut, and he is looking for aliens. Back when he was on earth, Jeff wanted to learn more about where he could find aliens. But he was really busy preparing for his trip, and by the time Jeff had made everything ready to go, he had to leave right away and he had no time to do research on aliens. Because he couldn't get any research done, Jeff thought he should ask his friend the scientist who lives in space about it. Jeff went to visit the scientist at his space station, and Jeff took all of the scientist's advice and decided where he should go. He said to the scientist, "I'm gonna go find aliens on Mars!" The scientist agreed that that was an excellent plan, and Jeff headed off on his adventure. On his way to Mars, he dropped by the moon in case there were aliens there too. He looked everywhere on the moon, but he didn't find any aliens, so he continued toward Mars. When he arrived on Mars, Jeff finally found aliens! They were very friendly, and Jeff was happy that he got to meet nice aliens.

- | | |
|-----------------------|--|
| Filler (Exp1-5): | Why did Jeff drop by the moon on his way to Mars? |
| Ambiguous (Exp2): | Where did Jeff tell someone that he was gonna see aliens? |
| Filled-gap (Exp2): | Where did Jeff tell someone at the space station that he was gonna see aliens? |
| Filled-gap, definite: | Where did Jeff tell the scientist at the space station that he was gonna see aliens? |

Item 3

It was a beautiful day in spring, so Lizzie decided she was going to go catch some butterflies in the park. Her Mom and Dad weren't home, so Lizzie thought she should tell her brother or sister about going to the park, so that Mom and Dad will know where she is when they get back. She first went to her brother's room, but he was taking a nap, and she couldn't tell him about catching butterflies. Instead, Lizzie looked for her sister. She looked all over the house but didn't see her sister anywhere! When she was about to give up, Lizzie heard her sister's voice in the basement! She went to the basement and said to her sister: "I'm gonna catch butterflies in the park!" Then, on her way to the park, Lizzie passed by a parking lot and saw a butterfly near it! She walked slowly toward the butterfly, but before Lizzie could get there, another girl came along and caught the butterfly! Lizzie didn't see any more butterflies there, so she kept walking toward the park. There were lots and lots of butterflies in the park, and she caught one in a jar and took it home with her. She liked the one that she caught, but she wished she could have caught more butterflies.

- Ambiguous (Exp1): Where did Lizzie explain that she was gonna catch butterflies?
- Filled-gap (Exp1): Where did Lizzie explain in the basement that she was gonna catch butterflies?
- Ambiguous (Exp2): Where did Lizzie tell someone that she was gonna catch butterflies?
- Filled-gap (Exp2): Where did Lizzie tell someone in the basement that she was gonna catch butterflies?
- Filled-gap, definite: Where did Lizzie tell her sister in the basement that she was gonna catch butterflies?

Item 5

Annie was taking horseback riding lessons. Today she got to go on a long ride all by herself. First she tried to ride up a mountain, but the mountain trail was too steep for the horse so she couldn't ride the horse there at all. Instead, Annie and her horse went riding in the forest. It was so much fun to ride the horse in the forest—along the way, Annie and her horse even got to jump over a log that was blocking the path! Annie was really proud of herself for riding all alone in the forest, so she wanted to tell her riding teacher about her great ride. But her teacher was busy teaching another student, so she couldn't talk to her. That night, Annie and her friend went camping at a campsite. Annie's friend was taking horseback riding lessons too, so he asked Annie about her ride. Annie said to him, "I rode a horse in the forest today, and we jumped over a big log!" Annie's friend was jealous that she's already allowed to go riding by herself.

- Ambiguous (Exp1): Where did Annie explain that she rode her horse?
- Filled-gap (Exp1): Where did Annie explain at the campsite that she rode her horse?
- Ambiguous (Exp2): Where did Annie tell someone that she rode her horse?
- Filled-gap (Exp2): Where did Annie tell someone at the campsite that she rode her horse?
- Filled-gap, definite: Where did Annie tell her friend at the campsite that she rode her horse?

Item 7

Emily likes to play outdoors. One day she was swinging on the swings, and she jumped off the swings from really high up! She balanced herself really well on the landing, so she didn't fall. When Emily got bored with the swings, she decided to go climb a really tall tree. She got up pretty high, but suddenly one of the branches broke, so she fell off the tree and hurt herself! But Emily was a brave girl, so she got back up right away and didn't cry at all. Emily wanted to tell her friends how pleased she felt about not crying after hurting herself, so she went to find some of her friends at the library. She found a friend of hers there, but the librarian came out and told her that in the library they must be very quiet! So, Emily couldn't talk to her friend and felt disappointed. But she had a good idea: She could go to the swimming pool to see more friends because at the pool she can talk as much as she wants! When she got to the pool and found her friends, she said to them, "I hurt myself falling out of the tree but didn't cry at all!" Emily was happy she could finally tell someone about her day, and her friends were impressed by how brave she was.

- Ambiguous (Exp1): Where did Emily explain that she hurt herself?
- Filled-gap (Exp1): Where did Emily explain at the pool that she hurt herself?
- Ambiguous (Exp2): Where did Emily tell someone that she hurt herself?
- Filled-gap (Exp2): Where did Emily tell someone at the pool that she hurt herself?
- Filled-gap, definite: Where did Emily tell her friends at the pool that she hurt herself?

Filler Stories:**Item 2**

Bill and his Mom went shopping yesterday and bought a new soccer ball, so Bill wanted to try it out after school today. At school, he wanted to tell his friends about playing soccer later, to see if some of them want to play with him. But he didn't want the teacher to get angry at him for talking in class, so he didn't say anything at school. After school, Bill knew that his Mom would be worried if he didn't get home till late, so he went home to tell her about his plans. He said to his mom: "I'm gonna go play soccer on the soccer field!" His mom was happy that he was already using the new soccer ball. On his way to the soccer field, he passed by a playground and saw some friends there. He tried to play soccer with them, but they were already flying kites, so they didn't want to play. But Bill's best friend decided to leave the playground and

go to the soccer field with Bill. They played soccer together in the soccer field, and Bill scored lots of goals with his new soccer ball.

- Filler: Why didn't Bill talk to his friends in class?
 Ambiguous (Exp1): Where did Bill explain that he was gonna go play soccer?
 Filled-gap (Exp1): Where did Bill explain at home that he was gonna go play soccer?
 Filled-gap (Exp5): Where did Bill tell his Mom at home that he was gonna go play soccer?

Item 4

Hannah watched a movie about treasure hunting one day, and she thought it would be fun to go in search of treasure herself. But first, she needed advice on where to find it. She went to ask a professor at the university about the best places to find treasure, but he had gone on vacation so she couldn't talk to him. Then she decided that the guide at the museum might know where to find treasure too, so she went to the museum and talked to him. The museum guide there told her all that he knew about where to find treasure, and Hannah said to him: "Thanks! I think I'll go find treasure in a shipwreck!" The guide agreed that that was a good place and told her about a shipwreck nearby. On her way to the shipwreck, Hannah passed a cave. She thought there might be treasure in the cave too, so she decided to go in and have a look around. But the cave was really, really dark, and Hannah couldn't see anything! Hannah got really scared, so she left the cave right away and kept going toward the shipwreck. When she made it to the sea, she put on her goggles, dove underwater, and went inside the shipwreck. She looked around really carefully and finally found a chest full of treasure! Hannah was really excited, and thought she should give some to the museum guide because he helped her find it.

- Filler: Why did Hannah leave the dark cave right away?

Item 6

Heidi really likes airplanes and always wanted to sit in a pilot's seat on an airplane. One day when she was at the airport with her family, she decided to sneak inside one of the airplanes to see the pilot's seat, but before she could get to the plane, a police officer stopped her because it was too dangerous. Heidi was really sad that she didn't get to see the inside of an airplane, so she went to an airplane museum in the neighborhood. At the airplane museum, there was an old airplane, and they let Heidi go sit in the pilot's seat! She felt like she was a real pilot, and Heidi liked it so much that she decided to be a pilot when she grows up. Heidi went home feeling really happy, and she wanted to tell her parents about her new dream of becoming a pilot. But they might worry that it's a dangerous job, so Heidi decided not to tell them about her dream. Instead, she went to school the next day and told her favorite science teacher about her plan. She said to her teacher, "I sat in a pilot's seat at the museum, and now I want to be a pilot!" The teacher was really happy for Heidi and taught her about things she needs to learn to become a pilot.

- Filler: Why couldn't Heidi ride the airplane at the airport?

Item 8

John had a long day at school and he was really hungry. When he got home, John found a piece of cake sitting in his room! He remembered that his Mom and grandma had made a cake together this morning. Perfect! But when he went to eat the cake, he tripped over a toy and dropped it on the floor! What a mess! John couldn't eat the cake now, and he was still hungry, so he went to look for the rest of the cake to get another piece. He found the cake in the kitchen and ate a piece there. The recipe that his Mom and grandma used turned out really well, and the cake was really delicious! Once he finished the cake, he went to the laundry room to tell his Mom about how great the cake tasted, but he could not talk to her because she was on the phone. John waited for a little while, but Mom kept talking on the phone and didn't pay any attention to him. He was disappointed that he couldn't talk to his Mom about how good the cake was. But John remembered that his grandma helped his Mom to make the cake, so he decided to go to his grandma's house next door to tell her how wonderful the cake was! John walked over to grandma's house and said to his grandma: "I dropped the piece of cake in my room, but then I ate a piece of the cake in the kitchen, and it tasted great!" John was really glad he could let her know how well the recipe worked, and John's grandma was very happy with herself.

- Filler: Why couldn't John talk to his mom?